



المعهد التكنولوجي العالي بمدينة العاشر من رمضان

Higher Technological Institute, Tenth of Ramadan City

Graduation Project

Smart Respiratory Support System (SRSS)

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18 August 2026
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DECLARATION STATEMENT

We, the undersigned students, hereby declare that the work presented in this project report entitled “Smart Respiratory Support System” is entirely our own original work.

We confirm that this project has been developed solely by us as part of our graduation requirements. All system design concepts, hardware implementation, software development, mobile application interface, and documentation have been carried out by the project team. Any external sources, references, or materials used in this work have been properly acknowledged and referenced.

We fully understand that any form of plagiarism or failure to properly credit sources constitute a violation of academic integrity and may result in disciplinary action.

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ABSTRACT

This project presents the design and implementation of a **Smart Respiratory Support System** based on an ESP32 microcontroller for real-time physiological monitoring and respiratory assistance. The proposed system is designed to monitor essential physiological and respiratory parameters, including blood oxygen saturation (SpO_2), heart rate, airflow rate, and temperature, using integrated sensors.

The collected sensor data are processed by the ESP32 microcontroller, where signal processing, data validation, and threshold-based monitoring algorithms are applied to evaluate the patient's condition in real time. Based on the measured values, the system generates appropriate control signals to manage the connected respiratory support components and ensure safe operation.

To enhance reliability and user safety, the system includes real-time data visualization through LCD displays, as well as visual and audible alarm indicators to notify users during abnormal conditions. In addition, a mobile application interface is developed to enable direct monitoring and control of the system through Wi-Fi communication between the smartphone and the ESP32 controller without relying on cloud services.

The developed system provides a low-cost, portable, and efficient solution for respiratory support applications. It aims to improve patient monitoring, enhance accessibility, and provide a practical embedded system approach suitable for home healthcare and assisted respiratory care environments. The proposed design focuses on reliability, real-time response, safety, and ease of use.

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NOMENCLATURES

Table 1: Nomenclature

Symbol / Term	Description	Unit
SpO₂	Blood oxygen saturation	%
HR	Heart rate	BPM
Flow	Airflow rate	L/min
Temp	Temperature	°C
SD	Sample standard deviation	Same unit as the measured parameter
CV	Coefficient of variation	%
P1-P5	Participant identifiers used in the reported measurements	-
t	Time or duration	s or ms
f	Pulse or signal frequency	Hz or kHz
V	Electrical voltage	V
λ	Ultraviolet wavelength	nm
MAX3010X	Optical sensor family used for SpO ₂ and heart-rate measurement	-
MAX30102	Pulse-oximetry and heart-rate sensor used in participant measurements	-
MAX30105	Optical sensor model referenced during firmware initialization	-
DS18B20	Digital temperature-sensor probe	-
YF-S401	Pulse-output airflow sensor	-
0x26 / 0x27	I ² C addresses assigned to the two LCD modules	Hexadecimal address

ACRONYMS

Table 2: Acronyms

Acronym	Meaning	Acronym	Meaning
AC	Alternating Current	LED	Light-Emitting Diode
ADC	Analog-to-Digital Converter	OLED	Organic Light-Emitting Diode
BPM	Beats Per Minute	PCB	Printed Circuit Board
COVID-19	Coronavirus Disease 2019	PPG	Photoplethysmography
CV	Coefficient of Variation	QR	Quick Response
DNA	Deoxyribonucleic Acid	RNA	Ribonucleic Acid
ECG	Electrocardiogram	SDG	Sustainable Development Goal
ESP32	Espressif 32-bit microcontroller platform	SPI	Serial Peripheral Interface
GPIO	General-Purpose Input/Output	SRSS	Smart Respiratory Support System
HTTP	Hypertext Transfer Protocol	SSID	Service Set Identifier
I²C	Inter-Integrated Circuit	UART	Universal Asynchronous Receiver/Transmitter
IP	Internet Protocol	USB	Universal Serial Bus
IoT	Internet of Things	UV-C	Ultraviolet C
JSON	JavaScript Object Notation	VOC(s)	Volatile Organic Compound(s)
LCD	Liquid Crystal Display	Wi-Fi	Wireless networking technology

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Chapter 1: Introduction

1.1 Background of the Study

Respiratory diseases and oxygenation-related complications represent a significant global healthcare challenge and remain among the major causes of morbidity and mortality worldwide. Blood oxygen saturation (SpO_2) is one of the most important physiological parameters used to evaluate respiratory efficiency and overall patient condition. A decrease in SpO_2 levels may indicate respiratory distress, impaired lung function, or insufficient oxygen supply to body tissues. Therefore, continuous monitoring of respiratory-related parameters is essential for patients suffering from chronic respiratory diseases, sleep-related breathing disorders, post-operative complications, and elderly individuals requiring continuous healthcare support.

Conventional respiratory support systems, including mechanical ventilators and oxygen therapy equipment, are mainly used in hospitals and intensive care units. Although these systems provide effective respiratory assistance, they are generally expensive, require professional supervision, and are not always suitable for home healthcare environments. In many situations, patient monitoring depends on periodic measurements performed by healthcare providers or caregivers, which may delay the detection of sudden changes in the patient's physiological condition.

Recent developments in embedded systems have enabled the design of compact and cost-effective healthcare devices capable of performing real-time physiological monitoring and control. Microcontrollers such as the ESP32 provide powerful processing capabilities, integrated Wi-Fi communication, low power consumption, and suitable interfaces for connecting biomedical sensors and control components. Furthermore, biomedical sensors such as pulse oximeters, flow sensors, and temperature sensors allow continuous measurement of important respiratory and physiological parameters.

By integrating biomedical sensing, embedded processing, wireless communication, and control mechanisms, modern respiratory support systems can provide continuous monitoring and faster response to abnormal conditions. Such systems can improve patient safety, support home healthcare applications, and provide a practical alternative for developing affordable respiratory assistance solutions.

1.2 Project Definition

The **Smart Respiratory Support System** is an ESP32-based embedded healthcare system designed to monitor and support respiratory conditions through real-time measurement, processing, and control. The system continuously acquires important physiological and respiratory parameters, including blood oxygen saturation (SpO_2), heart rate, airflow rate, and temperature, using integrated sensors.

The collected sensor data are processed by the ESP32 microcontroller, which applies signal processing, data validation, and predefined threshold-based monitoring logic to evaluate the patient's condition. According to the measured values, the system generates appropriate control signals to operate connected respiratory support components and maintain safe operation.

The system provides real-time visualization of patient information through LCD displays and includes safety notification features using LED indicators and audible alarms during abnormal conditions. In addition, a mobile application interface is developed to allow direct monitoring and control of the system through Wi-Fi communication between the smartphone and the ESP32 controller without depending on cloud services or external platforms.

The project aims to provide a low-cost, portable, and practical respiratory support solution suitable for educational development, research applications, and assisted healthcare environments.

1.3 Problem Statement

Respiratory complications and sudden changes in oxygen levels can become life-threatening if they are not detected and addressed promptly. Existing respiratory support systems are typically expensive, complex, and primarily designed for hospitals and intensive care environments, which limits their availability and suitability for home healthcare applications.

Furthermore, many conventional monitoring systems focus mainly on measuring physiological parameters and generating alerts without providing an integrated control mechanism for respiratory assistance. This may result in delayed response during critical situations, especially for patients who require continuous monitoring and rapid intervention.

The absence of affordable, portable, and integrated respiratory support systems creates a need for a solution capable of continuously monitoring essential physiological parameters, evaluating patient status in real time, providing appropriate respiratory assistance, and allowing convenient supervision through a user-friendly mobile interface.

The proposed Smart Respiratory Support System addresses these challenges by combining biomedical sensors, an ESP32-based control unit, respiratory support components, safety mechanisms, and a mobile application interface into a unified embedded system.

1.4 Motivation

The development of the Smart Respiratory Support System is motivated by the increasing demand for accessible and cost-effective healthcare technologies that can support patients with respiratory difficulties. Continuous monitoring of parameters such as SpO₂, heart rate, airflow rate, and temperature play an important role in detecting changes in patient condition and improving response time.

Traditional respiratory monitoring and support equipment are often expensive and require specialized medical environments and professional operation. This creates limitations for home-

based healthcare and small-scale medical applications. Advances in embedded systems provide new opportunities to develop compact, reliable, and affordable solutions capable of performing real-time monitoring and control.

The proposed system aims to bridge the gap between basic physiological monitoring devices and complete respiratory support solutions by integrating sensor acquisition, embedded processing, automated control, safety features, and direct mobile application communication through Wi-Fi.

Through this integration, the system seeks to improve monitoring efficiency, enhance user accessibility, and demonstrate the potential of embedded technologies in healthcare applications.

1.5 Aim of the Project

The aim of this project is to design and implement a **Smart Respiratory Support System based on an ESP32 embedded platform** capable of continuously monitoring important physiological and respiratory parameters, including SpO₂, heart rate, airflow rate, and temperature, while providing controlled respiratory assistance when abnormal conditions are detected.

The system is designed to enhance patient safety through real-time monitoring, threshold-based control logic, alarm mechanisms, LCD visualization, and direct communication with a mobile application using Wi-Fi technology. The proposed design provides a practical, low-cost, and portable solution for respiratory support and healthcare assistance applications.

1.6 Objectives of the Project

The objectives of the Smart Respiratory Support System are divided into technical and non-technical objectives. These objectives focus on developing a reliable embedded healthcare system capable of monitoring respiratory-related parameters, providing controlled assistance, and improving accessibility of respiratory care solutions.

1.6.1 Technical Objectives

The main technical objectives of this project are:

- Design and implement a real-time physiological monitoring system capable of measuring important patient parameters, including blood oxygen saturation (SpO₂), heart rate, airflow rate, and temperature using integrated sensors.
- Develop an ESP32-based embedded control system responsible for acquiring sensor data, processing measurements, and applying threshold-based monitoring logic to evaluate patient conditions.

- Implement control mechanisms for respiratory support components using output interfaces such as relay modules and solenoid valves according to predefined safety conditions.
- Develop a mobile application interface that enables direct monitoring and control of the system through Wi-Fi communication between the smartphone and ESP32 controller.
- Display real-time patient information, including SpO₂, heart rate, flow rate, temperature, operating mode, and system status through LCD displays.
- Implement multiple operating modes to support different system functions, including Monitoring Mode, Nebulizer Mode, Ventilation Mode, Controlled Ventilation Mode, Emergency Mode, and Standby Mode.
- Integrate safety features including visual indicators, audible alarms, emergency shutdown functionality, manual control options, and sensor validation mechanisms.
- Evaluate system performance by testing sensor readings, response time, communication reliability, control accuracy, and overall system operation under different conditions.

1.6.2 Non-Technical Objectives

The non-technical objectives of the project include:

- Enhance patient safety by providing continuous monitoring and rapid response to abnormal respiratory conditions.

(SDG 3 – Good Health and Well-being)

- Support the development of accessible healthcare solutions that can assist patients outside traditional hospital environments.

(SDG 3 & SDG 10 – Reduced Inequalities)

- Provide a cost-effective prototype solution that demonstrates the potential of embedded systems in healthcare applications.

(SDG 9 – Industry, Innovation, and Infrastructure)

- Improve the usability of respiratory support systems through a simple mobile application interface that allows convenient monitoring and control.

(SDG 3 – Good Health and Well-being)

- Encourage innovation in medical engineering by integrating biomedical sensors, embedded controllers, and wireless communication technologies.

(SDG 9 – Industry, Innovation, and Infrastructure)

- Support educational and research activities related to smart healthcare devices and embedded medical systems.

(SDG 4 – Quality Education)

1.7 System Overview

The Smart Respiratory Support System is an ESP32-based embedded healthcare prototype designed to monitor patient physiological conditions and provide controlled respiratory assistance when required. The system integrates sensing, processing, control, communication, and safety functions into a unified platform.

The sensing unit consists of biomedical sensors responsible for measuring essential parameters, including blood oxygen saturation (SpO₂), heart rate, airflow rate, and temperature. The collected measurements are transmitted to the ESP32 microcontroller, which serves as the main processing and control unit of the system.

The ESP32 processes the received sensor data using filtering, validation, and threshold-based monitoring logic to determine the current system condition. According to the evaluated measurements, the controller generates appropriate control signals to operate connected respiratory support components.

The system provides real-time information display through LCD screens, showing measured physiological parameters, operating mode, and system status. In addition, LED indicators and a buzzer are integrated to provide visual and audible alerts during abnormal or emergency conditions.

A mobile application interface is developed to communicate directly with the ESP32 through Wi-Fi connection. The application allows users to monitor patient parameters, select operating modes, and control system functions without requiring cloud platforms or external servers.

The proposed system is designed to provide a low-cost, portable, and practical embedded solution for respiratory monitoring and assisted respiratory care applications.

1.8 Scope of the Project

The scope of this project includes the design, implementation, and evaluation of an ESP32-based Smart Respiratory Support System prototype. The system is designed to:

- Monitor blood oxygen saturation (SpO₂), heart rate, airflow rate, and temperature continuously in real time.
- Acquire and process sensor measurements using an ESP32 microcontroller.
- Apply threshold-based monitoring logic to evaluate patient status and detect abnormal conditions.
- Control respiratory support components through output interfaces based on system requirements.
- Display real-time physiological measurements, operating modes, and system status using LCD displays.
- Provide visual and audible alerts using LEDs and a buzzer during critical conditions.
- Enable direct communication between the mobile application and ESP32 through Wi-Fi.
- Provide user control through different operating modes, including Monitoring, Nebulizer, Ventilation, Controlled Ventilation, Emergency, and Standby modes.
- Incorporate safety mechanisms such as manual override, sensor validation, and emergency shutdown.
- Support educational, research, and prototype healthcare applications.

Limitations

The proposed system is developed as a prototype for respiratory monitoring and assisted support applications. It is not intended to replace certified medical ventilators, intensive care equipment, or professional medical supervision. The system should not be considered a life-support device and requires further clinical validation before any medical deployment.

1.9 Significance of the Project

The Smart Respiratory Support System contributes to healthcare engineering by integrating biomedical sensing, embedded control, wireless communication, and user interaction into a single platform. The significance of the proposed system can be summarized as follows:

- Provides an affordable embedded solution for monitoring important respiratory and physiological parameters.
- Improves patient safety by enabling continuous monitoring and rapid detection of abnormal conditions.
- Supports home healthcare applications by reducing dependence on continuous hospital-based observation.
- Demonstrates the practical integration of biomedical sensors with ESP32-based embedded systems.
- Provides controlled respiratory assistance through automated hardware control mechanisms.

- Enhances system accessibility through direct mobile application communication using Wi-Fi technology.
- Encourages the development of portable healthcare devices suitable for educational and research purposes.
- Demonstrates the role of embedded technologies in improving healthcare monitoring and patient support systems.

1.10 System Architecture

The **Smart Respiratory Support System** follows a modular embedded system architecture that integrates sensing, processing, control, communication, user interface, and safety mechanisms. The architecture is designed to provide continuous monitoring of respiratory-related parameters and enable controlled operation of respiratory support components through an ESP32-based control unit.

The system is divided into several functional layers, where each layer performs a specific task while communicating with other layers to achieve reliable real-time operation.

1.10.1 Sensor Layer

The sensor layer is responsible for collecting real-time physiological and respiratory measurements from the patient and the respiratory system.

The sensing layer includes:

• SpO₂ Sensor

- Measures blood oxygen saturation levels.
- Provides information about the patient's oxygenation condition.

• Heart Rate Measurement

- Monitors the patient's heart rate.
- Helps evaluate the patient's physiological status.

• Flow Rate Sensor

- Measures the airflow rate within the respiratory pathway.
- Provides feedback about air delivery performance.

- **Temperature Sensor**

- Measures temperature values required for system monitoring.

The collected sensor data are continuously transferred to the ESP32 microcontroller for processing and evaluation

1.10.2 Processing Layer

The processing layer is implemented using the **ESP32 microcontroller**, which represents the main control unit of the system.

The ESP32 is responsible for:

- Acquiring data from connected sensors.
- Processing and validating sensor measurements.
- Applying filtering techniques to improve measurement reliability.
- Comparing measured values with predefined safety thresholds.
- Executing control logic according to the selected operating mode.
- Managing communication with the mobile application through Wi-Fi.
- Controlling output devices and safety mechanisms.

The use of ESP32 provides sufficient processing capability, integrated wireless communication, and flexible interfacing with different hardware components.

1.10.3 Actuation Layer

The actuation layer is responsible for controlling the respiratory support components according to the decisions generated by the ESP32 controller.

The actuation layer includes:

- Relay module.
- Solenoid valve.
- Air delivery control components.

The ESP32 generates control signals that operate the relay module, which controls connected actuators to regulate airflow and provide respiratory assistance according to predefined system conditions.

The actuation process is performed according to threshold-based control logic to ensure safe and reliable operation.

1.10.4 Interface Layer

The interface layer provides communication between the system and the user, allowing monitoring and control of system functions.

The interface layer consists of:

- **LCD Displays**
 - Display real-time values including:
 - SpO₂ level.
 - Heart rate.
 - Flow rate.
 - Temperature.
 - Operating mode.
 - System status.
- **LED Indicators**
 - Provide visual indication of system states and warnings.
- **Buzzer**
 - Generates audible alarms during abnormal or emergency conditions.
- **Mobile Application**
 - Provides a user-friendly interface for monitoring and controlling the system.
 - Communicates directly with the ESP32 through Wi-Fi.
 - Allows selection of operating modes and viewing system parameters.

1.10.5 Safety Layer

The safety layer is designed to improve system reliability and protect against abnormal operating conditions.

Safety mechanisms include:

- Sensor validation to detect invalid or unreliable measurements.
- Threshold-based alarm generation.
- Emergency shutdown functionality.

- Manual control and override options.
- Fail-safe operating logic.
- Continuous system status monitoring.

These mechanisms ensure that the system can respond safely during sensor failures, abnormal readings, or emergency situations.

1.10.6 Workflow Overview

The overall operation of the proposed system follows the sequence below:

1. The integrated sensors continuously measure the patient's physiological and respiratory parameters, including SpO₂, heart rate, flow rate, and temperature.
2. The collected sensor data are transmitted to the ESP32 microcontroller.
3. The ESP32 processes the incoming measurements and performs data validation and filtering.
4. The processed values are compared with predefined safety thresholds according to the selected operating mode.
5. Based on the evaluated conditions, the ESP32 generates control signals to operate the connected respiratory support components.
6. The LCD displays real-time measurements, operating mode, and system status.
7. Visual indicators and audible alarms are activated when abnormal or emergency conditions are detected.
8. The mobile application communicates directly with the ESP32 through Wi-Fi to monitor system parameters and control available functions.
9. The system continuously repeats this process to ensure real-time monitoring, safe operation, and rapid response. As shown in Figure 1.1, the Smart Respiratory Support System combines sensing, ESP32-based processing, actuation, user interfaces, and safety functions within a continuous monitoring workflow.

Smart Respiratory Support System Workflow

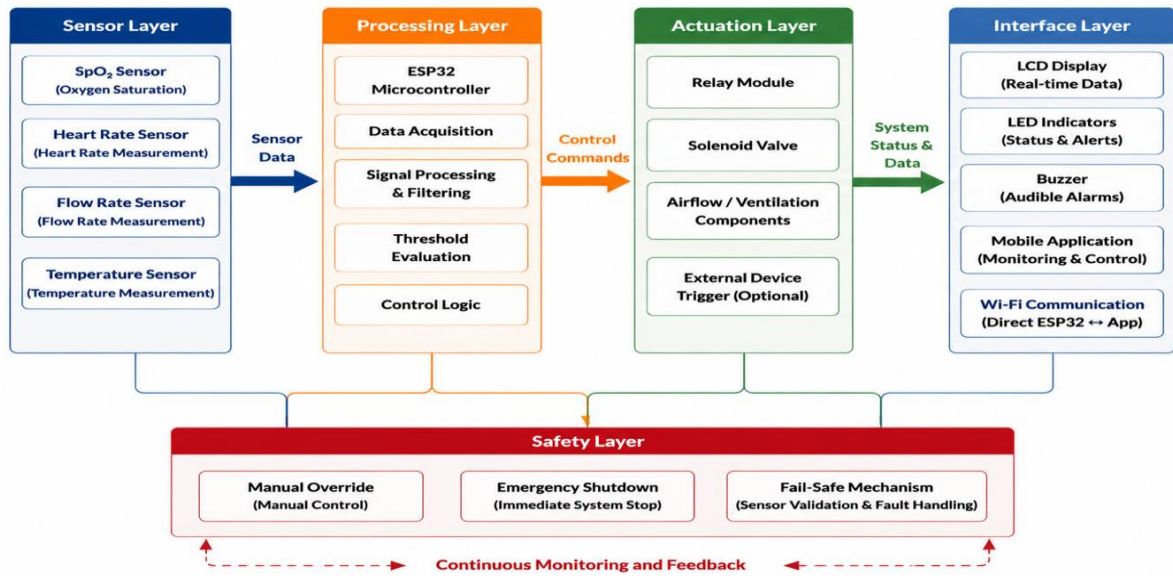


Figure 1.1: Smart Respiratory Support System Workflow

1.10.7 System Design Considerations

The proposed system is designed according to several engineering requirements, including:

- Real-time data acquisition and processing.
- Low response time.
- Reliable sensor integration.
- Modular hardware architecture.
- Easy maintenance and future expansion.
- Low-cost implementation.
- Safe operation during continuous monitoring.

The modular design allows individual components such as sensors, communication interfaces, and actuators to be modified or upgraded without affecting the complete system operation.

1.10.8 Real-Time Performance Requirements

Real-time performance is a critical requirement because the system continuously monitors patient-related parameters and must respond quickly to abnormal conditions.

The ESP32 microcontroller provides adequate processing capability to perform:

- Sensor data acquisition.
- Data processing and validation.
- Threshold comparison.
- Control signal generation.
- Wi-Fi communication with the mobile application.

The system is designed to minimize response delay and provide timely alerts and control actions during critical situations.

1.10.9 Reliability and Fault Tolerance

To achieve reliable operation, the system incorporates several fault-handling mechanisms, including:

- Sensor validation.
- Continuous system monitoring.
- Fail-safe control logic.
- Emergency shutdown functionality.
- Alarm notification during abnormal conditions.

If incorrect sensor readings or abnormal system behaviour are detected, the controller initiates appropriate safety actions and alerts the user through the available notification methods.

1.10.10 Future Enhancements

Future improvements of the Smart Respiratory Support System may include:

- Development of advanced Android and iOS mobile applications.
- Integration of additional biomedical sensors such as ECG and blood pressure sensors.
- Improvement of hardware design for clinical-level applications.
- Enhancement of communication reliability and user interface features.
- Integration of additional safety and monitoring functions.

1.11 Data-Driven Analysis of Respiratory Disease Burden

The following section presents a statistical analysis of the global burden of respiratory diseases using data-driven visualization. The aim is to highlight the severity of respiratory-related mortality, regional differences, and vulnerable population groups in order to justify the need for affordable and reliable respiratory monitoring and support systems.

Charts showing global and African respiratory disease deaths from 1990 to 2023, including the COVID-19 spike in 2020.

As shown in Figure 1.2, the chart presents the proportional distribution of the main respiratory-related causes of death.

Chronic respiratory diseases, lower respiratory infections, and COVID-19 represent major contributors to respiratory mortality, while other causes account for a smaller proportion. The visualization highlights the importance of continuous respiratory monitoring and early detection of abnormal conditions.

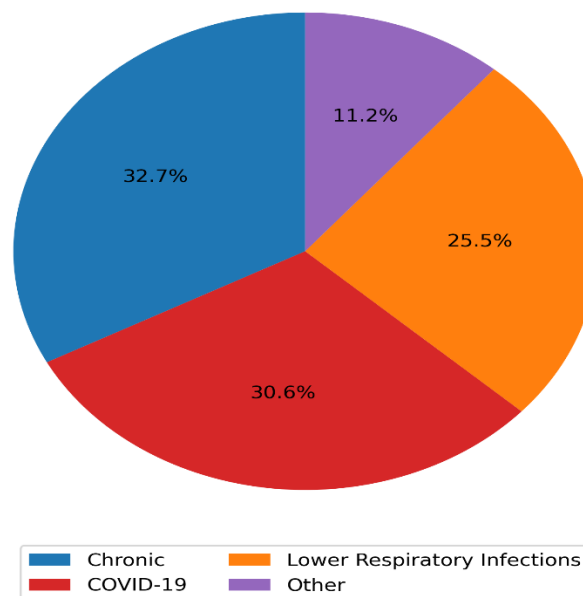


Figure 1.2: Respiratory Disease Causes Distribution

As shown in Figure 1.3, the chart illustrates changes in the major contributors to global respiratory mortality from 1990 to 2023, including the marked COVID-19 increase in 2020. The presented trends demonstrate the continuing impact of chronic respiratory diseases and lower respiratory infections worldwide and emphasize the importance of developing effective monitoring and support solutions.

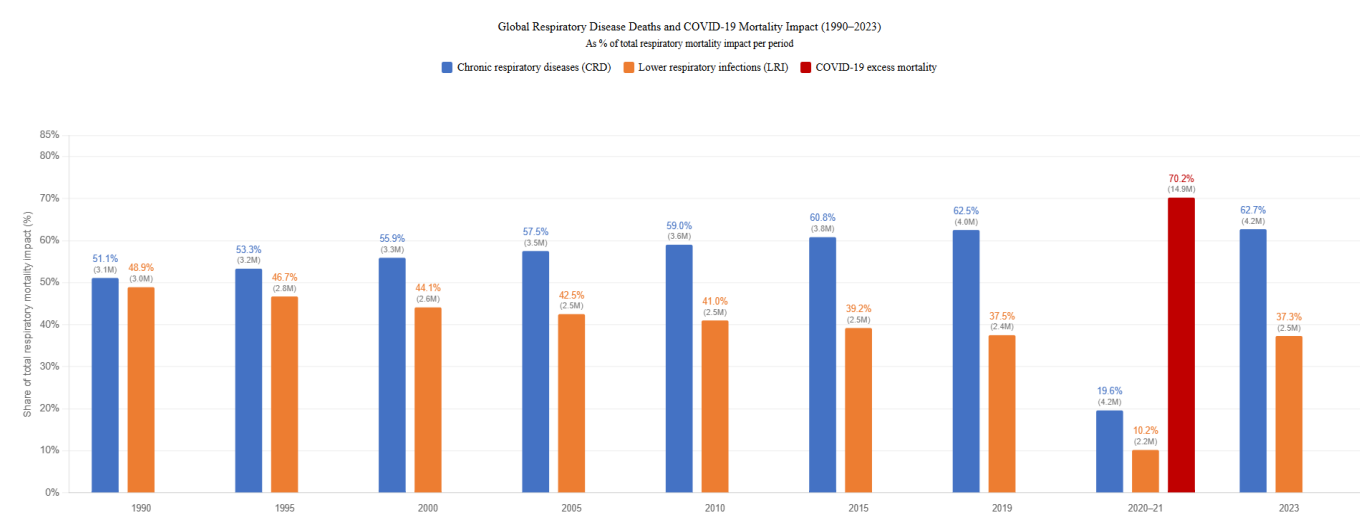


Figure 1.3 :Global Respiratory Disease Mortality Trend (1990–2023)

As shown in Figure 1.4, the chart compares the respiratory disease burden in Africa with the corresponding global values across the selected years. The comparison highlights regional differences and demonstrates the need for accessible healthcare technologies, especially in areas where continuous medical monitoring resources may be limited.

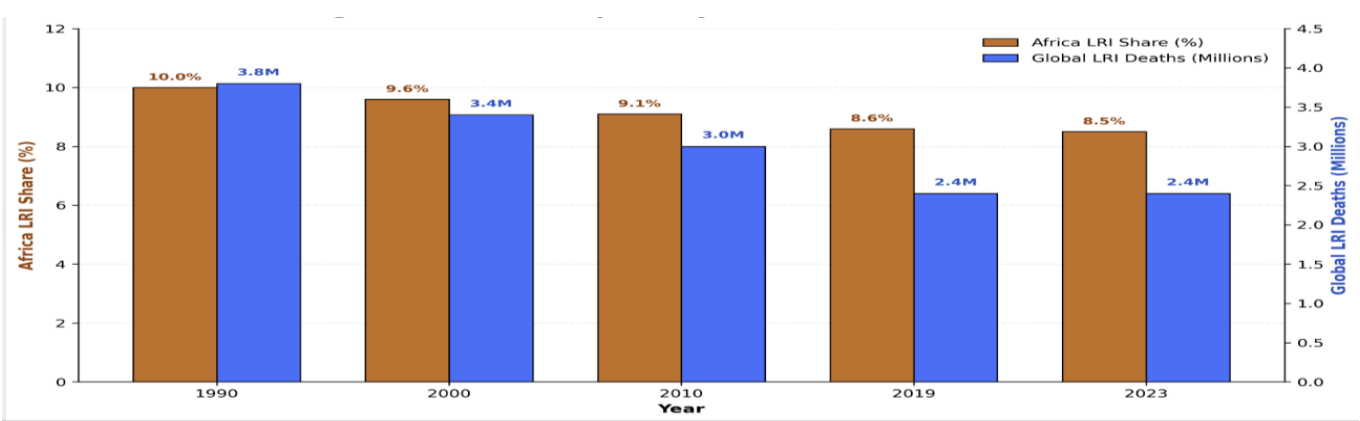


Figure 1.4:Regional Comparison of Respiratory Disease Burden (Africa vs Global

As shown in Figure 1.5, the chart presents the share of lower respiratory infection deaths among children under five in Africa compared with the rest of the world.

The results emphasize the vulnerability of young populations and the importance of developing affordable respiratory monitoring and support technologies.

■ Africa share of child LRI deaths ■ Rest of world



Figure 1.5: Distribution of Lower Respiratory Infection Deaths in Children Under 5 Years

Chapter 2: Literature Review and Related Work

2.1 Overview of Respiratory Monitoring and Support Systems

Respiratory monitoring and support systems play an essential role in modern healthcare by providing continuous observation of patient conditions and assisting individuals suffering from respiratory disorders. Among the most important physiological parameters used for respiratory assessment are blood oxygen saturation (SpO_2), heart rate, airflow rate, and temperature.

Traditional respiratory support equipment, such as hospital ventilators, provides effective respiratory assistance but is usually expensive, complex, and dependent on professional operation. These limitations create the need for affordable, portable, and user-friendly embedded systems capable of monitoring respiratory parameters and providing controlled support when required.

Recent developments in embedded systems have enabled the integration of biomedical sensors, microcontrollers, wireless communication, and control components into compact healthcare devices. Microcontrollers such as ESP32 provide sufficient processing capability and built-in Wi-Fi communication, making them suitable for real-time monitoring applications and direct interaction with mobile applications.

Previous research has mainly focused on physiological monitoring and wireless data transmission, while fewer studies have addressed the integration of real-time monitoring with automated respiratory support mechanisms. The proposed Smart Respiratory Support System aims to address this gap by combining sensor monitoring, embedded processing, control mechanisms, safety features, and mobile application interaction in a single platform.

2.2 ESP32-Based Health Monitoring System

Manjhi et al. presented an ESP32-based patient monitoring system designed to measure vital parameters such as heart rate, temperature, and SpO_2 . The system utilizes biomedical sensors connected to an ESP32 microcontroller and uses IoT communication through the Blynk platform for remote visualization of collected data.

The study demonstrated the effectiveness of ESP32 and pulse oximetry sensors for continuous physiological monitoring. However, the system mainly focuses on data acquisition and visualization without providing automated respiratory assistance, actuator control, or safety mechanisms for critical conditions.

2.3 Portable Pulse Oximeter System for SpO₂ Monitoring

Kakumanu et al. developed a portable pulse oximeter system for measuring heart rate and SpO₂ levels. The system incorporates an OLED display for real-time visualization and provides wireless transmission capabilities for accessing measured parameters.

This work highlights the importance of compact and portable physiological monitoring devices. However, the proposed system is limited to measurement and data presentation, without implementing automated control actions, respiratory support components, or integrated safety responses.

2.4 ESP32-Based Wireless Patient Monitoring System

Rahman et al. proposed an ESP32-based wireless patient monitoring system capable of measuring multiple physiological parameters, including heart rate, oxygen saturation, and ECG signals.

The system demonstrates the ability of embedded controllers to collect and transmit healthcare data efficiently. However, the system does not include automated actuation, respiratory assistance mechanisms, or control logic capable of responding to abnormal patient conditions.

2.5 Advanced SpO₂-Based Respiratory Condition Detection

Hoang and Liang investigated the use of continuous SpO₂ signals for detecting respiratory abnormalities, particularly sleep apnea conditions. Their study analysed physiological patterns obtained from wearable SpO₂ measurements to improve respiratory condition assessment.

The research highlights the importance of continuous oxygen saturation monitoring in identifying respiratory problems, especially in home healthcare environments. The study demonstrates that SpO₂ signals can provide valuable information about respiratory conditions and patient status.

However, the proposed approach mainly focuses on respiratory condition detection and analysis without including real-time control mechanisms, respiratory support components, or automated intervention. Therefore, it does not provide immediate assistance when abnormal conditions occur.

Unlike this approach, the proposed Smart Respiratory Support System focuses on real-time threshold-based monitoring and embedded control using an ESP32 microcontroller to evaluate patient status and operate connected respiratory support components when required.

2.6 Technical Comparison of Related Works

A comparison between the reviewed studies and the proposed Smart Respiratory Support System is presented in the following tables. The comparison focuses on the main technical aspects related to physiological monitoring, system implementation, control capabilities, and safety features.

As shown in Table 2.1, the reviewed systems are compared in terms of monitored parameters, controller type, display method, and mobile-interface capability.

Table 2.1: Comparison of Monitoring and Processing Features

Technical Feature	Manjhi et al.	Kakumanu et al.	Rahman et al.	Hoang & Liang	Proposed Project
SpO₂ Monitoring	Yes	Yes	Yes	Yes	Yes
Heart Rate Monitoring	Yes	Yes	Yes	Yes	Yes
Flow Rate Measurement	No	No	No	No	Yes
Temperature Measurement	Yes	No	No	No	Yes
ESP32 Controller	Yes	No	Yes	No	Yes
LCD/OLED Display	Yes	Yes	Yes	No	Yes
Mobile Application Interface	Through Platform	Web Interface	Remote Interface	No	Direct Wi-Fi App

As shown in Table 2.2, the control and safety capabilities of the reviewed systems are compared with those included in the proposed project design.

Table 2.2: Comparison of Control and Safety Features

Technical Feature	Manjhi et al.	Kakumanu et al.	Rahman et al.	Hoang & Liang	Proposed Project
Threshold-Based Monitoring	Limited	No	Limited	No	Yes
Advanced Data Analysis	No	No	No	Yes	No
Automated Respiratory Support	No	No	No	No	Yes
Relay/Solenoid Control	No	No	No	No	Yes
Alarm System	Limited	No	Limited	No	Yes
Safety Mechanisms	Limited	No	No	No	Yes
Operating Modes	No	No	No	No	Yes

Analysis of Comparison:

The comparison shows that most previous studies successfully implemented physiological monitoring using embedded systems and biomedical sensors. However, these systems mainly focused on data acquisition, visualization, and communication without providing integrated respiratory support control.

The reviewed works also lack important features such as airflow monitoring, actuator-based respiratory assistance, safety mechanisms, and multiple operating modes.

The proposed Smart Respiratory Support System combines real-time monitoring of SpO₂, heart rate, flow rate, and temperature with ESP32-based processing, controlled respiratory support, alarm mechanisms, and direct mobile application communication through Wi-Fi. This integration provides a more complete embedded solution for respiratory monitoring and assisted support applications.

2.7 Low-Cost Embedded Respiratory Support Systems

Several studies have investigated the development of low-cost respiratory support devices using embedded controllers and electronic control components. These systems aim to provide affordable alternatives to conventional respiratory equipment by integrating sensors, controllers, and actuators into compact designs.

Embedded respiratory support systems commonly utilize microcontrollers to manage sensing, control operations, and user interaction. Components such as valves, airflow control mechanisms, and monitoring sensors can be combined to provide controlled respiratory assistance according to predefined operating conditions.

However, many existing systems focus either on respiratory assistance or physiological monitoring separately. This separation limits their ability to provide a complete solution that combines continuous patient monitoring, controlled support, safety mechanisms, and user-friendly interaction.

The proposed Smart Respiratory Support System addresses this limitation by integrating biomedical sensors, an ESP32 controller, respiratory support components, alarm mechanisms, and a mobile application interface into a unified embedded platform.

2.8 Analysis of Related Works

The reviewed studies demonstrate the effectiveness of embedded systems and biomedical sensors in healthcare monitoring applications.

Manjhi et al., Kakumanu et al., and Rahman et al. successfully developed systems capable of measuring important physiological parameters such as SpO₂ and heart rate. These studies confirm the suitability of microcontrollers and biomedical sensors for continuous patient monitoring. However, their systems mainly focus on data acquisition and visualization without providing respiratory assistance or hardware control mechanisms.

Hoang and Liang demonstrated the importance of continuous SpO₂ analysis for identifying respiratory abnormalities. However, their work focuses mainly on assessment and detection rather than real-time intervention or respiratory support.

Compared with previous research, the proposed Smart Respiratory Support System provides an integrated approach by combining:

- Real-time physiological monitoring.
- ESP32-based embedded processing.
- Threshold-based condition evaluation.
- Controlled respiratory support operation.
- Alarm and safety mechanisms.
- Direct Wi-Fi communication with a mobile application.

This integration creates a practical embedded solution capable of monitoring patient conditions and providing controlled respiratory assistance when required.

2.9 Research Gap Identification

The reviewed studies demonstrate significant progress in the development of embedded healthcare systems for physiological monitoring and respiratory condition assessment. Most existing systems successfully utilize biomedical sensors and microcontrollers to measure important parameters such as SpO₂ and heart rate.

However, several limitations remain in previous research. Many existing solutions focus mainly on data acquisition, visualization, or wireless transmission without integrating real-time respiratory support control. In addition, most systems do not combine multiple respiratory parameters such as oxygen saturation, heart rate, airflow rate, and temperature within a single embedded platform.

Furthermore, previous monitoring systems often lack integrated safety mechanisms, actuator control, and user-friendly interfaces that allow direct interaction with the system. These limitations reduce their ability to provide rapid responses during abnormal respiratory conditions.

The proposed **Smart Respiratory Support System** addresses these gaps by integrating biomedical sensors, an ESP32-based control unit, respiratory support components, alarm mechanisms, safety features, and a mobile application interface through direct Wi-Fi communication. The system applies threshold-based monitoring and control logic to provide real-time evaluation and controlled respiratory assistance while maintaining a simple and cost-effective embedded design.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the methodology used for the design and implementation of the **Smart Respiratory Support System**. The proposed system is based on an ESP32 microcontroller that integrates biomedical sensing, airflow control, user interface, and safety mechanisms into a single embedded platform.

The system is designed to monitor important physiological and respiratory parameters, including blood oxygen saturation (SpO_2), heart rate, airflow rate, and temperature. The collected measurements are processed by the ESP32 controller, where threshold-based monitoring logic is applied to evaluate the system condition and control the connected respiratory support components.

The methodology describes the complete system design, including the hardware configuration, software structure, operating modes, communication method, and safety features implemented in the prototype. The system consists of an air supply and delivery unit, sensor subsystem, embedded control unit, output interfaces, and a mobile application interface connected through direct Wi-Fi communication.

The developed system is considered a research and educational prototype. It demonstrates the integration of embedded systems and biomedical sensors for respiratory monitoring and assisted support applications. It is not intended to replace certified medical ventilators or clinical respiratory equipment and requires further medical validation before practical healthcare use.

3.2 System Architecture Overview

The architecture of the **Smart Respiratory Support System** consists of five main functional subsystems:

1. **Air Supply and Treatment Unit**
2. **Air Delivery and Regulation Unit**
3. **Monitoring and Sensor Unit**
4. **Embedded Control Unit**
5. **User Interface and Communication Unit**

These subsystems operate together under the control of the ESP32 microcontroller to provide real-time monitoring, controlled respiratory assistance, and safe system operation.

As shown in Table 3.1, the system is divided into five cooperating subsystems, each with a defined function and a corresponding set of key components.

Table 3.1: Functional Subsystems of the Smart Respiratory Support System

Subsystem	Function	Key Components
Air Supply & Treatment	Provides and prepares the airflow source before delivery.	Compressor, Air Supply Components, Filtration Elements
Air Delivery & Regulation	Controls airflow delivery according to system operation.	Solenoid Valve, Flow Control Element
Monitoring System	Measures patient and system parameters in real time.	SpO ₂ Sensor, Heart Rate Sensor, Flow Rate Sensor, Temperature Sensor
Embedded Control	Processes sensor data and manages system operation.	ESP32, Relay Module, Control Logic
User Interface & Communication	Provides monitoring and control functions.	LCD Display, Buzzer, Mobile Application, Wi-Fi Communication

3.2.1 Air Pipeline

The air path is arranged as a sequential hardware pipeline. Ambient air is pressurized by the compressor, stored and stabilized in the tank, cleaned from oil and water contamination, passed through a carbon filter to reduce selected contaminants and odors, exposed to UV-C light for microbial reduction, and finally controlled by a solenoid valve before reaching the patient interface.

As shown in Figure 3.1, the air-treatment path consists of six sequential stages extending from ambient-air intake to controlled delivery at the patient interface.



Figure 3.1: Six-stage air treatment and delivery pipeline

3.2.2 Full System Block Architecture

ESP32 acts as the central controller of the system. It receives sensor feedback, processes the measured data, executes the selected operating mode, and controls the connected output components through the relay interface.

The sensor subsystem provides measurements of SpO₂, heart rate, flow rate, and temperature. The ESP32 evaluates these parameters using threshold-based monitoring logic and generates control signals for the solenoid valve and alarm components.

The system provides local feedback through the LCD display and buzzer. In addition, the mobile application communicates directly with the ESP32 through Wi-Fi to monitor system parameters and control available operating modes.

As shown in Figure 3.2, the complete system architecture links the sensor subsystem, ESP32 controller, airflow regulation, local output devices, and the mobile application.

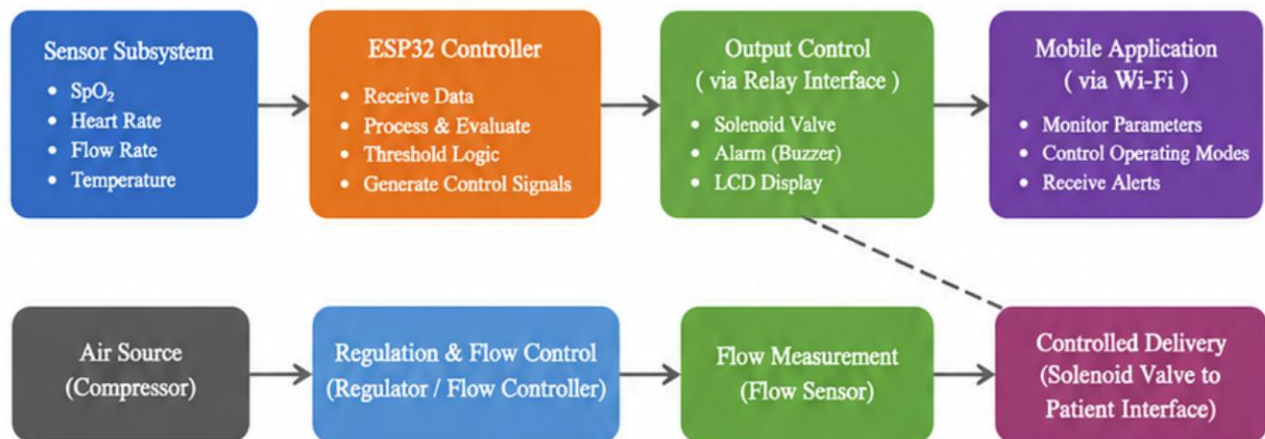


Figure 3.2: Control, sensing, actuation, and mobile application architecture

3.3 Hardware Tools

This section describes the hardware components used in the Smart Respiratory Support System prototype. The hardware is organized according to its functional role in the system, including:

- Air supply and delivery components.
- Biomedical and system sensors.
- Embedded controller.
- Output and safety components.

The integration of these components enables real-time monitoring, controlled airflow delivery, and safe system operation.

3.3.1 Air Supply and Treatment Pipeline

The air supply and treatment section provides the airflow source required for respiratory support operation. This subsystem is responsible for generating and preparing airflow before it reaches the delivery stage.

The main components of this subsystem include:

As shown in Table 3.2, the air-supply and treatment pipeline is organized into six successive stages extending from air compression to controlled delivery.

Table 3.2: Air Supply and delivery stages

Stage	Component	Role in the Prototype
1	Compressor	Draws ambient air and raises its pressure to create a usable air supply for the downstream pipeline.
2	Air Tank	Stores compressed air and acted as a pressure buffer to reduce pollution from the compressor.
3	Oil/Water Separator	Removes compressor oil mist and condensed water droplets before air reaches filters, sensors, and valves.
4	Carbon Filter	Reduces Odors and selected chemical contaminants; improves the air quality before sterilization and delivery.
5	UV-C Stage	Provides microbial reduction inside a sealed exposure chamber. It supports sterilization but does not guarantee hospital-grade sterility.
6	Solenoid Valve	Electrically opens and closes the delivery path under ESP32 control through an isolated relay interface.

Compressor and Air Tank

As shown in Figure 3.3, the compressor unit combines the pump head, motor, storage tank, pressure-control components, and outlet accessories required to generate and handle the compressed-air supply.



Figure 3.3: Air Compressor Unit and Storage Tank Unit

1. Air Compressor Pump Head

The air compressor pump head draws ambient air through the intake path and increases its pressure inside the compression chamber. The compressed air is then discharged toward the storage tank, where it becomes available for the downstream treatment stages. Its operating condition directly affects pressure build-up time, available airflow, and the overall response of the air-supply subsystem.

2. Motor

The motor provides the mechanical power required to drive the compressor pump head by converting electrical energy into rotational motion. Stable motor operation supports consistent compression and reliable pressure generation. The motor should operate within its rated supply and duty conditions because excessive loading or overheating can reduce performance and shorten component life.

3. Air Intake Filter

The air intake filter forms the first protection stage in the pipeline by reducing dust and relatively large airborne particles before they enter the compressor. This helps protect the

pump surfaces and prevents avoidable contamination of the downstream pneumatic path. Periodic inspection is important because a blocked intake filter can restrict airflow, increase motor load, and slow pressure build-up.

4. Check Valve

The check valve allows compressed air to move from the pump toward the storage tank while preventing reverse flow when the compressor stops. By isolating the tank from the pump head, it helps preserve stored pressure and reduces reverse mechanical loading during restart. A leaking or stuck check valve may cause pressure loss, repeated compressor cycling, or delayed system response.

5. Air Tank (15L Storage Unit)

The 15 L air tank acts as a compressed-air reservoir and pressure buffer between the compressor and the delivery pipeline. It smooths short-term pressure pulsations and separates the compressor operating cycle from the immediate airflow demand of the downstream system. The tank also provides a temporary stored supply, but its pressure must remain within the permitted operating limits and accumulated condensate should be managed during maintenance.

6. Pressure Gauge

The pressure gauge provides a direct visual indication of the pressure stored inside the tank. It allows the operator to confirm that pressure is building normally, identify unexpected pressure loss, and verify the condition before opening the downstream path. The gauge is a monitoring device rather than a control element, so its readability and accuracy are important for safe adjustment and troubleshooting.

7. Pressure Switch (Automatic ON/OFF Control)

The pressure switch automatically starts and stops the compressor according to predefined lower and upper tank-pressure limits. This control prevents continuous motor operation, maintains a usable pressure reserve, and reduces the risk of excessive pressure build-up. Its settings must be coordinated with the regulator and the mechanical safety devices because an incorrect or faulty switch can cause excessive cycling or unsafe operating conditions.

8. Outlet Valve

The outlet valve provides manual isolation and controlled release of compressed air from the tank toward the treatment and regulation stages. It is used during startup, shutdown, adjustment, and maintenance to separate the stored-pressure section from the downstream

pipeline. Opening the valve gradually helps avoid sudden pressure changes and allows the operator to check fittings and connections before full operation.

9. Control Panel

The control panel groups the compressor ON/OFF controls and the available operating indicators in one accessible location. It allows the operator to energize or stop the air-supply unit and to observe its basic operating state before enabling the respiratory-support pipeline. Clear labeling and easy access to the controls support safer startup, manual intervention, and routine troubleshooting.

10. Rubber Foot (Vibration Isolation)

The rubber feet reduce the transfer of compressor vibration to the bench or enclosure and help limit movement during operation. Lower vibration improves mechanical stability, reduces noise, and decreases stress on pneumatic fittings and electrical connections. The feet should remain evenly positioned and should be inspected for wear because damaged supports may allow the unit to shift or transmit excessive vibration.

Oil/Water Separator

As shown in Figure 3.4, the air-pressure regulator stabilizes the compressed air supplied from the tank, while the integrated gauge, adjustment knob, and moisture-separation bowl support controlled downstream delivery.

Compressed air can carry condensed moisture and residual aerosols from the supply path. Inside the separator, the internal airflow path encourages droplets and heavier contamination to collect in the lower bowl, while the regulator adjusts the pressure delivered to the next stage. The bowl should be inspected and drained when necessary, and this component should be treated as a conditioning stage rather than a substitute for validated medical-grade air purification.



Figure 3.4: Air Pressure Regulator with Moisture Separator

Carbon Filter

As shown in Figure 3.5, the carbon filter is an air-treatment component that uses activated-carbon media to reduce odors, volatile organic compounds (VOCs), and selected chemical impurities in the compressed airflow.

This stage absorbs fine contaminants that may pass through the earlier treatment stages, providing an additional purification layer before airflow delivery.

The effectiveness of activated carbon depends on the number of media, airflow rate, contact time, and the degree of media saturation. Because its adsorption capacity is limited, the cartridge requires periodic inspection and replacement according to the operating conditions.

The carbon stage does not remove all moisture or microorganisms and therefore operates as one part of the complete treatment pipeline rather than as a standalone sterilization device.



Figure 3.5: Carbon Filter

UV-C Stage

As shown in Figure 3.6, the UV-C stage exposes the compressed airflow to germicidal ultraviolet radiation at approximately 254 nm inside a sealed treatment chamber.

This stage provides an additional microbial-reduction step after oil separation and carbon filtration. The UV-C light is intended to disrupt the DNA or RNA of microorganisms and reduce their ability to replicate.

The chamber is designed as a sealed enclosure to provide controlled airflow exposure while limiting direct ultraviolet-radiation leakage to the surrounding environment.

The microbial-reduction performance of the UV-C stage depends on radiation intensity, exposure time, airflow velocity, and chamber geometry. A sealed housing is essential because direct UV-C exposure can be harmful to the eyes and skin. For this prototype, the stage demonstrates the treatment concept; hospital-grade sterilization cannot be claimed without dose measurement, leakage testing, and microbiological validation.



Figure 3.6: UV-C Sterilization Chamber

Solenoid Valve

As shown in Figure 3.7, the solenoid valve is an electromechanical actuator that provides controlled opening and closing of the airflow path through electrical commands from the ESP32 relay interface.

The valve supports timed airflow delivery according to the selected operating mode and provides the mechanical switching required for ventilation cycles.

When its coil is energized, the generated magnetic field moves the internal plunger and changes the valve state. When the electrical command is removed, the valve returns to its default condition according to its mechanical design.

The selected valve must be compatible with the operating pressure, airflow direction, electrical supply, and required response time. Leak-tight fittings and a defined safe default state are important, while the flow sensor provides independent feedback that airflow actually changed after a command.



Figure 3.7: Solenoid Valve Unit

3.3.2 Air Delivery and Regulation Unit

The air delivery and regulation unit is responsible for controlling and delivering the airflow generated by the air supply system. This subsystem ensures that the airflow reaches the output interface in a controlled and measurable manner according to the selected operating mode.

The pressure regulation and flow control components are used to maintain stable airflow conditions. The flow sensor provides feedback by measuring the delivered airflow rate, while the solenoid valve acts as the main actuator responsible for controlling the airflow path.

The solenoid valve is controlled by the ESP32 microcontroller through a relay module. The relay provides electrical isolation between the low-voltage control circuit and the valve power circuit, allowing safe operation of the actuator.

The main components of the air delivery and regulation unit are:

As shown in Table 3.3, the air-delivery unit combines pressure regulation, flow adjustment, airflow measurement, actuator control, and mechanical safety protection.

Table 3.3: Air delivery and regulation components

Component	Type	Function
Pressure Regulator	Mechanical / Pneumatic	Reduces and stabilizes airflow pressure before delivery.
Flow Control Element	Mechanical Restrictor	Helps regulate airflow and maintain stable operation.
YF-S401 Flow Sensor	Digital Pulse Flow Sensor	Measures airflow rate and provides feedback to the ESP32.
Solenoid Valve	Electromechanical Actuator	Controls airflow opening and closing according to ESP32 commands through the relay module.
Safety Protection Element	Mechanical Safety	Provides additional protection against abnormal operating conditions.

Detailed Operation of the Regulation Components

The elements summarized in the table perform different but complementary functions. Pressure regulation establishes a safe working pressure, flow adjustment sets the delivered rate, sensing confirms the actual output, and the switching and protection elements control or limit the pneumatic path.

Pressure Regulator

The pressure regulator reduces the higher and variable tank pressure to a lower, more stable downstream working pressure. The adjustment should be performed while observing the outlet gauge and the actual flow condition because the displayed pressure may change when airflow begins. The regulator does not generate airflow; it controls the pressure made available by the upstream supply.

Flow Control Element

The flow-control element introduces an adjustable restriction in the delivery line. Increasing or decreasing this restriction changes the relationship between pressure and airflow and allows the output to be tuned for the required operating condition. Its adjustment should be gradual because sudden changes can produce unstable flow and affect the sensor reading.

YF-S401 Flow Sensor

Within the regulation unit, the YF-S401 acts as the feedback device that confirms the quantity of airflow passing through the delivery path. The sensor output must be calibrated for the actual tubing, fittings, pressure, and air medium used in the prototype. Correct flow direction, secure connections, and reduced turbulence near the sensor improve measurement repeatability.

Solenoid Valve

The solenoid valve functions as the final electrically controlled gate in the airflow path. The ESP32 issues the logical command through the relay, and the valve changes between its open and closed states according to the active operating mode. Comparing the command with the measured flow helps identify conditions such as a blocked line, a disconnected actuator, or a valve that did not change state.

Safety Protection Element

The mechanical safety element provides an additional layer of protection against abnormal pressure or reverse-flow conditions. Depending on the final assembly, this function may be provided by a relief device, check mechanism, or another rated protective component. Whenever

possible, this protection should operate independently of software so that a controller fault does not remove the basic pneumatic safeguard.

3.3.3 Prototype Components and Practical Integration

The prototype hardware consists of different components responsible for airflow control, measurement, switching, and system monitoring. These components are integrated together through the ESP32 controller to achieve real-time monitoring and controlled operation.

As shown in Table 3.4, the main prototype components are assigned specific roles in airflow control, measurement, electrical switching, and temperature monitoring.

Table 3.4: Prototype components and their roles in the system

Component	Specification	System Role
Solenoid Valve	AC-powered electromechanical valve	Main airflow actuator. It controls the air path and must be operated through a relay module.
YF-S401 Flow Sensor	Digital pulse output flow sensor	Measures airflow rate and provides feedback to the ESP32 controller.
Relay Module	Single-channel switching module	Provides electrical isolation and controls the solenoid valve safely.
Temperature Sensor Probe	Digital or analog temperature sensing element	Measures system temperature during operation.

Practical Integration Approach

The practical assembly combines pneumatic, low-voltage, and actuator-power connections in one prototype. The airflow components should be connected in the defined pipeline order, while the sensor wiring and ESP32 control wiring should be routed clearly so that each signal can be inspected and maintained.

Electrical and Pneumatic Separation

Low-voltage sensor and controller wiring should be separated from the solenoid-valve power path as far as practical. The ESP32 provides only logic-level control to the relay input, while the relay contacts switch the external actuator circuit. On the pneumatic side, fittings must be secured and checked for leakage, and each directional component must be installed according to the intended flow direction.

Feedback-Based Integration

The system does not rely only on the command sent to the actuator. The flow sensor supplies feedback that can be compared with the commanded valve state, while the temperature and physiological sensors provide the measurements used by the monitoring logic. This arrangement supports end-to-end observation of the command, physical response, and displayed system status.

Electrical Safety Consideration

The solenoid valve is an AC-powered component and must not be connected directly to the ESP32 GPIO pins.

The ESP32 controls only the input side of the relay module, while the relay switches the valve power circuit. Proper electrical isolation, protection, and safe wiring practices must be applied during system assembly.

Relay Module

As shown in Figure 3.8, the relay module acts as the switching interface between the low-voltage ESP32 control circuit and the external solenoid-valve power circuit.

In the proposed system, the ESP32 controls only the relay input, while the relay contacts switch the actuator supply and maintain electrical separation between both circuits.

This arrangement protects the ESP32 from the higher voltage and current levels associated with the external load.

The relay module contains a low-voltage control input and electrically separate switching contacts, commonly identified as common, normally open, and normally closed terminals. The selected contact arrangement determines the actuator default condition when the relay is not energized.

The relay input must be compatible with the ESP32 logic level, while the contact rating must exceed the voltage and current required by the solenoid valve. Safe wiring, secure terminals, and suitable protection are necessary because the relay provides a switching boundary but does not by itself correct incorrect high-voltage installation.



Figure 3.8: Relay Module

3.3.4 ESP32 Microcontroller

As shown in Figure 3.9, the ESP32 development board serves as the central processing and control unit of the Smart Respiratory Support System.

It receives sensor measurements, processes the collected data, executes the selected operating mode, manages Wi-Fi communication, and generates control signals for the system outputs.

The controller interfaces with different signal types within the prototype. The optical sensor and LCD modules use digital communication buses, the flow sensor provides pulse feedback, and the remaining inputs and outputs are connected through suitable GPIO interfaces.

During operation, the ESP32 repeatedly acquires measurements, checks their validity, evaluates the selected state, updates the local displays, and handles alarms or user commands. Organizing these operations in a repeated control loop allows monitoring and interface updates to continue without depending on an external computer.

The integrated Wi-Fi capability supports direct communication with the mobile interface. However, the GPIO pins are low-voltage logic outputs and must not drive the solenoid valve or other high-power loads directly; the relay and the external supply form the required interface to those components.

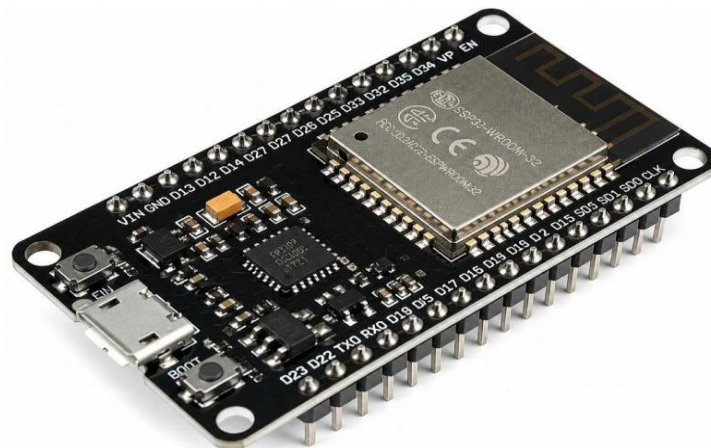


Figure 3.9: ESP32 Microcontroller

The main functions of the ESP32 include:

- Reading SpO₂ and heart rate values from the MAX3010X sensor.
- Reading airflow measurements from the YF-S401 flow sensor.
- Monitoring temperature values.
- Applying threshold-based monitoring logic.
- Controlling the relay module and solenoid valve.

- Updating the LCD display.
- Activating the buzzer during abnormal conditions.
- Communicating with the mobile application through Wi-Fi.

The ESP32 was selected because of its integrated Wi-Fi capability, sufficient processing performance, and compatibility with various sensors and electronic modules

As shown in Table 3.5, the selected ESP32 features support wireless communication, peripheral interfacing, sensor acquisition, and simultaneous monitoring and control.

Table 3.5: ESP32 features relevant to the implementation

ESP32 Feature	Relevance to Prototype
Integrated Wi-Fi	Enables direct communication with the mobile application.
Multiple GPIO Pins	Allows connection with relay, buzzer, LCD, and sensor interfaces.
ADC Channels	Supports analog-based sensors when required.
I²C / SPI / UART Communication	Enables connection with sensors and display modules.
Processing Capability	Allows simultaneous monitoring and control operations.

3.3.5 Sensor Subsystem

The sensor subsystem is responsible for collecting patient and system measurements required for real-time monitoring and control. The system uses biomedical and airflow sensors to provide continuous feedback to the ESP32 controller.

The measured parameters include:

- Blood oxygen saturation (SpO₂).
- Heart rate.
- Airflow rate.
- Temperature.

The collected sensor data are processed by the ESP32 and used for monitoring system status and executing threshold-based control decisions.

As shown in Table 3.6, the sensor subsystem measures SpO₂, heart rate, airflow rate, and temperature using the integrated sensing elements.

Table 3.6: Sensor subsystem summary

Sensor	Parameter	Unit	Role
MAX3010X	SpO ₂ and Heart Rate	% and bpm	Measures oxygen saturation and pulse rate for patient monitoring.
YF-S401 Flow Sensor	Airflow Rate	L/min	Measures delivered airflow and verified air delivery operation.
Temperature Sensor	Temperature	°C	Monitors system temperature for safety evaluation.

- **MAX3010X(SpO₂ and Heart Rate)**

As shown in Figure 3.10, the MAX3010X is an optical biomedical sensor used to obtain blood oxygen saturation (SpO₂) and heart-rate measurements through photoplethysmography (PPG).

The sensor uses red and infrared light with a photodetector to capture changes in blood volume during each heartbeat, and the collected signals are processed by the ESP32 to calculate the physiological values.

Communication with the ESP32 is provided through the I²C interface, supporting continuous and non-invasive acquisition of the measured parameters.

During measurement, stable finger placement and sufficient contact with the optical surface are important because motion, ambient light, weak perfusion, or incorrect positioning can distort the PPG waveform. The firmware should therefore reject unstable or physically unreasonable values instead of immediately treating every calculated value as valid.

Heart rate is obtained from the periodicity of the pulse waveform, while SpO₂ estimation uses the relative behavior of the red and infrared optical signals. The sensor is suitable for prototype monitoring, but accuracy depends on signal quality, algorithm implementation, and comparison with a calibrated reference instrument.

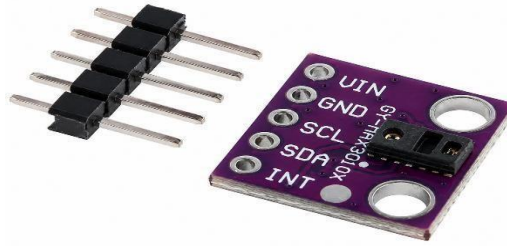


Figure 3.10: MAX3010X SpO2 and Heart Rate Sensor

- **YF-S401 Flow Sensor**

As shown in Figure 3.11, the YF-S401 flow sensor measures airflow by generating electrical pulses whose frequency is proportional to the rotational speed of its internal turbine.

The ESP32 processes these pulses to calculate the airflow rate in real time, allowing the system to verify delivery and detect conditions such as insufficient flow or valve-operation problems.

The controller counts the pulses over a defined sampling interval and converts the measured pulse frequency into a flow-rate value using a calibration relationship. Because the conversion can be affected by the airflow path and sensor installation, the calibration factor should be established using the complete assembled pipeline rather than assumed from an isolated sensor.

For repeatable operation, the sensor should be installed in the correct flow direction with secure, leak-free connections. Sharp bends, sudden restrictions, or strong turbulence close to the sensor can change the turbine response and should be minimized during assembly and testing.



Figure 3.11: YF-S401 Flow Sensor

• Temperature Probe

As shown in Figure 3.12, the temperature probe provides continuous temperature feedback to the ESP32 during system operation.

The measured value is processed by the controller and displayed through the user interface to support system-status monitoring and the detection of abnormal temperature conditions.

The position of the probe determines what the measurement represents, so it should be placed consistently at the selected monitoring point and protected from direct contact with unrelated heat sources. The metallic probe requires time to reach thermal equilibrium, which introduces a response delay compared with rapidly changing electrical signals.

The firmware can reduce false alarms by checking the reading range, handling disconnected-sensor values and applying suitable averaging before threshold comparison. Final temperature limits and calibration should be confirmed for the actual probe type and its installation location.



Figure 3.12.: Temperature Probe

3.3.6 Output and Interface Subsystem

The output and interface subsystem provides system feedback, user interaction, and actuator control. The outputs are controlled by the ESP32 according to the selected operating mode and measured sensor values.

As shown in Table 3.7, the output and interface subsystem combines local display, audible feedback, actuator switching, airflow control, and mobile communication.

Table 3.7: Output and interface subsystem

Output	Controlled By	Purpose
LCD Display	ESP32 via I ² C	Displays SpO ₂ , heart rate, flow rate, temperature, operating mode, and system status.
Buzzer	ESP32 GPIO	Provides audible alarms during abnormal conditions.
Relay Module	ESP32 GPIO	Controls the electrical switching of the solenoid valve.
Solenoid Valve	Relay Controlled Supply	Controls airflow delivery according to system commands.
Mobile Application	Wi-Fi Communication	Provides monitoring and operating mode selection.

LCD Displays

The local LCD displays provide immediate access to measured values and system status without requiring a phone connection. The ESP32 updates the screens through the I²C interface, allowing several display modules to share the communication bus when their addresses are configured correctly. Clear labels, stable refresh timing, and readable status messages help the user distinguish measurements from operating-mode or alarm information.

Buzzer

The buzzer provides an audible indication when the firmware detects an abnormal or emergency condition. It is driven from an ESP32 output through an appropriate interface and can be assigned different activation patterns for normal feedback and urgent alarms. The buzzer should be treated as one notification method and not as the only safety mechanism.

Relay and Solenoid Output Path

The actuator command follows a defined chain from the ESP32 logic output to the relay input and then through the relay contacts to the solenoid-valve supply. This arrangement keeps the controller separated from the actuator power circuit. The measured airflow can then be compared with the commanded valve state to determine whether the physical delivery path responded as expected.

Chapter 4: Results and Discussion

4.1 Introduction

This chapter presents a bench-level functional evaluation of the Smart Respiratory Support System (SRSS) prototype under normal indoor conditions. The evaluation covers system initialization, acquisition and presentation of the four monitored parameters—heart rate, oxygen saturation (SpO₂), airflow, and temperature—together with the local LCD interface, physical controls, Wi-Fi communication, mobile web application, operating-mode transitions, and emergency software interlock.

Sensor measurements were recorded during five adult-participant sessions after the displayed readings had stabilized. One synchronized measurement set was retained for each participant, producing five complete measurement sets and twenty recorded parameter values. The results were summarized using the mean, sample standard deviation, minimum, maximum, range, and coefficient of variation.

The recorded sensor values were not compared with certified clinical or laboratory reference instruments. Therefore, the presented results are interpreted as preliminary functional prototype measurements and descriptive bench-level findings rather than as evidence of clinical accuracy, calibration, or diagnostic suitability.

The emergency function evaluated in this chapter was limited to a software interlock consisting of the Stopped state, buzzer alarm, LCD warning, and physical acknowledgement requirement. Physical airflow interruption was outside the evaluated scope because the relay and solenoid actuation path was not controlled by the firmware built used during testing.

The chapter first describes the prototype, test arrangement, measurement conditions, and evaluation criteria. It then presents the initialization results, sensor outputs, local-interface operation, operating-mode transitions, wireless communication, mobile-application functions, and integrated system behavior. Finally, the findings are discussed in relation to the project objectives, implemented design, and principal prototype limitations.

4.2 Experimental Setup and Test Methodology

4.2.1 Prototype Under Test

All results in this chapter refer to the single bench-assembled SRSS prototype described in Chapter 3, operating with the firmware version used during the reported tests. The prototype integrates the ESP32 controller, three sensing modules that provide four monitored parameters, two I²C LCD modules, four mode-selection buttons, one STOP button, and a piezo buzzer. The MAX30102 optical module provides SpO₂ and heart-rate outputs, the DS18B20 probe provides temperature, and the YF-S401 sensor provides airflow.

As shown in Figure 4.1, the final SRSS prototype used throughout the experimental evaluation is presented from multiple external viewpoints.

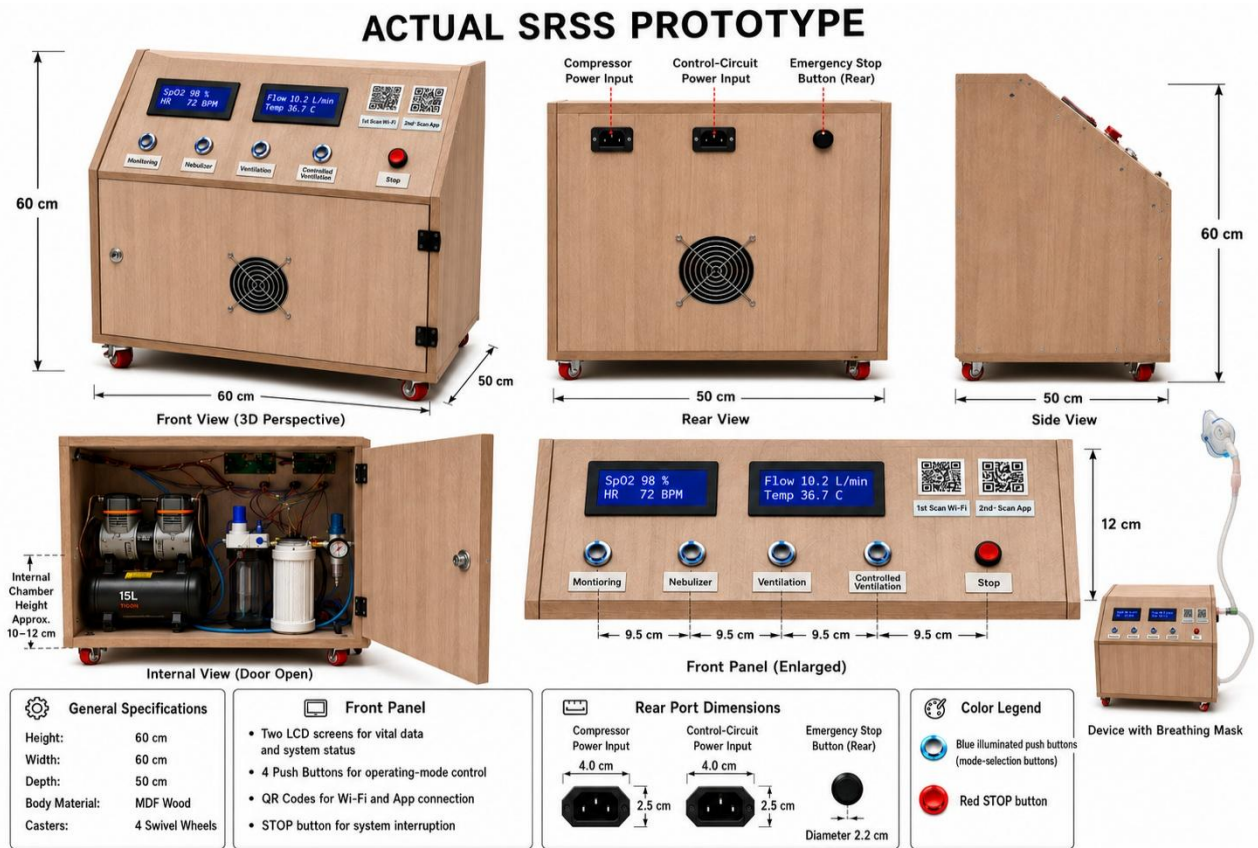


Figure 4.1: Actual SRSS Prototype Showing External Views, Internal Assembly, Front Panel, and Main Dimensions

Figure 19 presents the enclosure design from multiple viewpoints and identifies the dual LCD modules, mode-selection buttons, STOP button, power inputs, ventilation openings, and internal component arrangement.

4.2.2 Test Bench Arrangement

The prototype was positioned on a fixed indoor bench with the equipment required for operation and data recording. The arrangement included the SRSS prototype, a laptop for firmware uploading and Serial Monitor observation, a smartphone for accessing mobile web application, a regulated 5 V USB supply, and the airflow source and tubing. Each participant remained seated and at rest while the displayed outputs were allowed to stabilize before the synchronized measurement set was recorded.

The internal electronic arrangement of the prototype is shown in Figure 4.2. The rendered internal view identifies the principal low-voltage components and the airflow-delivery arrangement included in the prototype design.



Figure 4.2: Internal Component Arrangement of the SRSS Prototype

4.2.3 Test Equipment and Data-Collection Tools

The experimental evaluation was performed using the SRSS prototype together with the supporting equipment required for operation, observation, and data recording. No independent laboratory-certified or commercial reference instruments were used to compare the SpO_2 , heartrate, temperature, or airflow values produced by the prototype. The evaluation therefore focuses on output availability, consistency, descriptive variation, interface operation, and functional system behavior rather than measurement accuracy or calibration error.

The supporting equipment included a laptop for firmware uploading, Serial Monitor observation, and HTTP endpoint testing; a smartphone for accessing the mobile web application and scanning the QR codes; a regulated 5 V USB source for the low-voltage electronics; and the airflow source and tubing used with the YF-S401 sensor.

Table 4.1 summarizes the supporting equipment used to operate the prototype, observe its behavior, and record the experimental evidence.

Table 4.1: Test Equipment and Data-Collection Tools

Equipment	Purpose During Testing

Laptop	Uploading the firmware, monitoring Serial Monitor output and testing the HTTP endpoints.
Smartphone	Connecting to the ESP32 Wi-Fi network, accessing the mobile application, scanning the QR codes, and recording photographs and videos.
Regulated 5 V USB power source	Supplying power to the ESP32 and the connected low-voltage electronic components.
Air source, tubing, and breathing interface	Generating and routing airflow through the prototype airflow path and the YF-S401 sensor during functional testing.

4.2.4 Number of Trials

The number of measurements was deliberately limited, consistent with an initial graduation project evaluation rather than a clinical or metrological study. Heart rate, SpO₂, flow rate, and temperature were recorded from five adult volunteers at rest. One synchronized and stabilized four-parameter measurement set was retained for each participant, producing five complete measurement sets and twenty recorded parameter values.

The participant-level values were used to describe the distribution of the four prototype outputs across the tested group. Since only one stabilized set was retained for each participant, the analysis represents descriptive variation across participants rather than repeated-measurement repeatability for a single unchanged input.

System initialization was evaluated over ten power-up cycles. Each of the four active operating modes was selected and observed five times, resulting in twenty mode-selection trials. The STOP software interlock and the wireless/mobile-application path were each evaluated over ten operating cycles.

The selected measurements were sufficient to evaluate basic output availability, descriptive variation, interface behavior, mode transitions, and communication reliability at a preliminary prototype level. The limited participant sample and absence of calibrated reference instruments do not support clinical accuracy claims or formal agreement analysis.

Table 3.4 summarizes the participant measurements and repeated functional operating cycles completed during the prototype evaluation.

Table 4.3: Summary of Experimental Trials

Test	Participants / Conditions	Repetitions	Total Trials
Four-parameter sensor measurement	5 adult volunteers at rest	1 stabilized set per participant	5 sets / 20 values
System initialization	Power-up cycles	10 cycles	10 trials
Operating-mode selection	Four active modes	5 trials per mode	20 trials
STOP software interlock	Repeated activation	10 cycles	10 trials
Wireless and mobile application	Repeated operating cycles	10 cycles	10 trials

The participant measurements were treated as prototype outputs, while the remaining tests were evaluated according to functional success, interface response, and consistency of system behavior.

4.2.5 Pass/Fail Criteria

Pass/fail criteria were defined before interpreting the functional results in order to provide a consistent basis for evaluating the prototype. The criteria covered sensor-output availability, system initialization, operating-mode transitions, buzzer feedback, wireless communication, HTTP requests, and the emergency software interlock.

Because the sensor values were not compared with certified references, the sensor criterion assessed whether a synchronized four-parameter set became available, remained stable long enough to be recorded, and appeared consistently on the local and mobile interfaces. Table 17 summarizes the criteria applied during testing.

Table 4.4: Pass/Fail Criteria Applied During Prototype Testing

Test Category	Pass Criterion
System initialization	Both LCDs initialize, the required sensors are detected, and the system enters Monitoring without an unrecoverable error.
Sensor output	A stable heart rate, SpO ₂ , flowrate, and temperature set is displayed and recorded for each participant.
Mode transition by physical button	The selected operating mode is reflected in both LCDs and in the internal system state.
Mode transition by HTTP request	A valid request updates the internal state, both LCD displays, and the /status output.
Buzzer feedback	A short confirmation beep follows a valid mode change; the Stopped state produces the continuous alarm pattern.
Emergency software interlock	The system enters Stopped, activates the alarm and warning, and remains locked until physical Monitoring acknowledgement.
Wireless connection and HTTP endpoints	The phone connects, the local page loads, live values are displayed, and supported requests produce the expected response.

The criteria in Table 4.4 were used to classify each repeated functional trial as successful or unsuccessful. Sensor outputs were evaluated for availability and interface consistency, while system functions were evaluated according to the expected state transition, display response, alarm behavior, and communication response.

4.3 System Initialization and Firmware Verification

At power-up, the ESP32 firmware initiated the complete system start-up sequence. The firmware created the local Wi-Fi access point using the SSID “TiTo”, the password “12345678”, and the default IP address 192.168.4.1, then started the HTTP server. It also initialized the DS18B20 temperature sensor at 10-bit resolution using non-blocking conversion, configured the YF-S401 pulse interrupt on the falling edge, initialized the five push buttons and buzzer, and started both LCD modules.

The firmware also attempted to detect the MAX30105 sensor on the I²C bus. If the sensor was not detected during the first attempt, the system automatically repeated the detection procedure after a 3 s interval. This retry mechanism was included to improve start-up reliability without requiring manual intervention.

During the boot sequence, both LCDs displayed the message “SRSS booting...” before switching to the live monitoring screen. The initialization test was performed to verify that all

required firmware subsystems entered their expected operating states before normal monitoring began.

Table 4.5 summarizes the expected and observed behavior of each initialization step, including Wi-Fi and HTTP server start-up, sensor initialization, LCD boot indication, interrupt configuration, and the overall boot-to-ready time.

Table 4.5: System Initialization and Firmware Verification Results

Initialization Step	Expected Behavior	Observed Result
Wi-Fi access-point start-up	The “TiTo” access point becomes visible, and the address 192.168.4.1 is reachable.	Pass (10/10 power-up cycles). The access point became visible after 2.3 ± 0.3 s, and 192.168.4.1 was reachable in every cycle.
HTTP server start-up	The main routes (/ , /data, /status, and /mode) respond without error.	Pass (40/40 route checks: 4 routes $\times$ 10 cycles). All tested routes returned valid responses without an unrecoverable server error.
DS18B20 initialization	A valid temperature value becomes available within one request cycle.	Pass (10/10 cycles). The first valid temperature output appeared after 0.9 ± 0.2 s.
MAX30105 detection and auto-retry	The sensor is detected immediately or during the automatic retry after 3 s.	Pass (10/10 cycles). The sensor was detected immediately in 9 cycles and after one automatic 3 s retry in 1 cycle.
YF-S401 interrupt attachment	Pulse counting is enabled on the falling edge of the sensor signal.	Pass (10/10 airflow trials). The pulse counts increased when airflow was applied and returned close to zero after airflow was removed.
LCD1/LCD2 boot-message verification	Both displays show “SRSS booting...” during start-up before presenting live data.	Pass (20/20 display checks: 2 LCDs $\times$ 10 cycles). Both displays showed the boot message before switching to live output.
Boot-to-ready time	The elapsed time from power-on until the first valid live display is observed is recorded.	Mean boot-to-ready time: 5.2 ± 0.4 s. The measured range was 4.7–5.8 s across 10 power-up cycles.

The results in Table 18 show that the complete initialization sequence was successfully completed during all ten power-up cycles. The Wi-Fi network, HTTP server, sensors, interrupt-based airflow input, push buttons, buzzer, and LCD modules entered their expected operating states without an unrecoverable failure. The measured mean boot-to-ready time was 5.2 s, with individual values ranging from 4.7 to 5.8 s. These results indicate that the firmware provided a repeatable and reliable start-up sequence under the reported bench-test conditions.

4.4 Sensor Results

This section presents the four sensor outputs recorded from five adult volunteers. One stabilized measurement set was retained for each participant. The analysis describes the recorded values and their variation across the tested group; it does not establish clinical accuracy because no certified reference instruments were used.

4.4.1 SpO₂ and Heart Rate

The MAX30102 sensor was sampled continuously, and SpO₂ and heart-rate values were computed using the Maxim heart rate and oxygen-saturation algorithm. The heart-rate value was reported directly from the algorithm output without applying an additional fixed correction.

Each participant remained seated at rest while the fingertip was held steadily on the optical sensor. The synchronized measurement set was recorded after the displayed outputs stabilized and the local LCD values were checked against the mobile interface.

Table 4.6 summarizes the measured SpO₂, heart rate, airflow, and temperature outputs recorded simultaneously for five adult participants at rest. Each row represents one stabilized measurement set.

Table 4.6: Prototype SpO₂, Heart Rate, Airflow, and Temperature Outputs

Participant	Heart Rate (BPM)	SpO ₂ (%)	Flow Rate (L/min)	Temperature (°C)
P1	72	98	10.2	36.7
P2	76	97	11.0	36.8
P3	68	99	9.6	36.5
P4	81	98	12.1	36.9
P5	74	97	10.7	36.6
Mean	74.2	97.8	10.72	36.7

The recorded physiological and temperature outputs were within the expected resting or normal interface ranges used for the prototype. The airflow values remained within the configured prototype operating band. These results are functional prototype measurements and do not establish clinical accuracy. Figure 4.5 compares the recorded SpO₂ and heart-rate values.

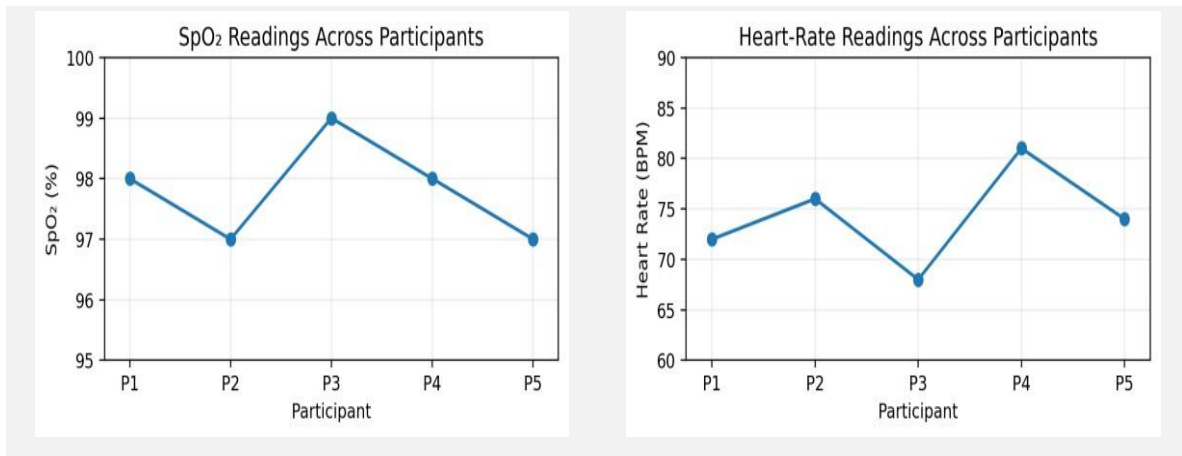


Figure 4.5: SpO₂ and Heart-Rate Readings Recorded Across the Five Participants

The SpO₂ values occupied a narrow 97–99% interval, whereas heart rate ranged from 68 to 81 BPM, reflecting normal participant-to-participant variation.

4.4.2 Temperature

The DS18B20 probe was configured for 10-bit resolution with non-blocking conversion. During each session, the probe was maintained in consistent skin contact, and the stabilized temperature output was recorded at the same time as the other three parameters. Figure 4.6 presents the temperature readings recorded across the five participant sessions.

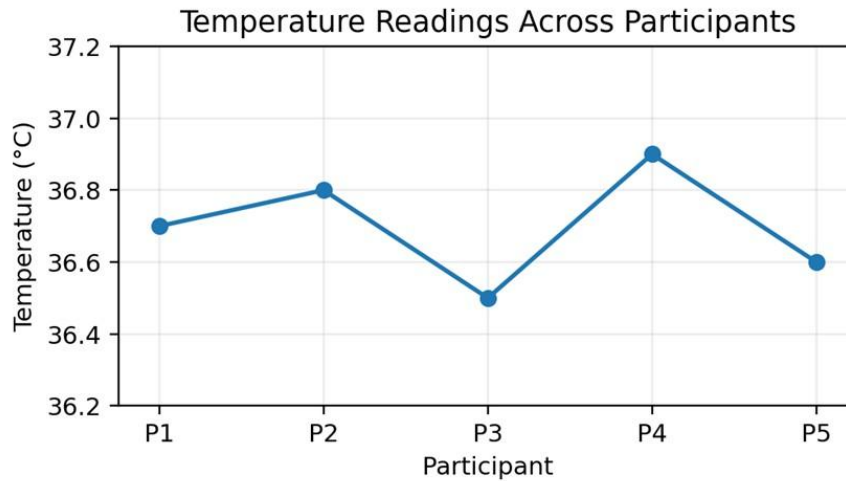


Figure 4.6: Temperature Readings Across Participants

The temperature readings ranged from 36.5 °C to 36.9 °C, with an overall mean of 36.7 °C and a total range of 0.4 °C.

The temperature values are also reported with the synchronized measurements in Table 4.6.

4.4.3 Airflow

Airflow was derived from pulse counting on GPIO 27 using a conversion factor of 98 pulses per liter. The corrected calculation used for reporting was $\text{Flow (L/min)} = (\text{pulses per second} \div 98) \times 60$, and the resulting value was presented directly on the LCD and through the /data endpoint without an additional division factor.

Because no calibrated reference flow meter was used, the airflow evaluation focused on the availability, stability, and distribution of the displayed prototype values across the five sessions. Figure 4.7 presents the recorded flow-rate values for the five participants.

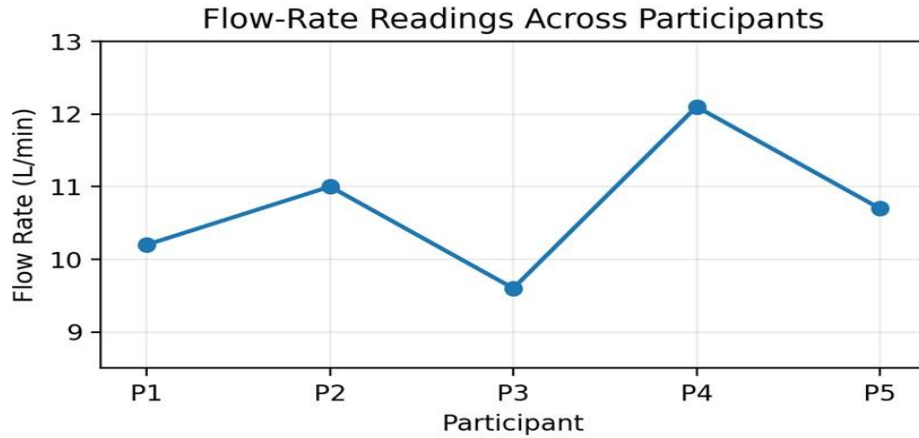


Figure 4.7: Flow-Rate Readings Across Participants

The flow-rate readings ranged from 9.6 to 12.1 L/min, with an overall mean of 10.72 L/min.

The recorded flow-rate values are also reported with synchronized measurements in Table 4.6. Their absolute accuracy remains unverified because the conversion was not compared with a calibrated reference flow meter.

4.4.4 Descriptive Variability Across Participants

Since one stabilized set was retained for each participant, the statistical analysis describes variation across the tested participants rather than repeated-measurement repeatability for one unchanged input. The summary includes the meaning, sample standard deviation, minimum, maximum, range, and coefficient of variation.

Table 4.7 presents the descriptive statistics calculated from the five recorded values for each parameter.

Table 4.7: Descriptive Statistics for the Five Participant Measurements

Parameter	Mean	(SD)	Minimum	Maximum	Range	CV (%)
SpO ₂ (%)	97.80	0.84	97.00	99.00	2.00	0.86
Heart Rate (BPM)	74.20	4.82	68.00	81.00	13.00	6.49
Temperature (°C)	36.70	0.16	36.50	36.90	0.40	0.43
Flow Rate (L/min)	10.72	0.94	9.60	12.10	2.50	8.74

The lowest relative variation was observed for temperature (CV = 0.43%) and SpO₂ (CV = 0.86%). Heart rate showed a CV of 6.49%, reflecting normal differences among participants, while flow rate showed the largest relative variation (CV = 8.74%). These statistics describe the recorded prototype outputs and do not quantify measurement accuracy.

Taken together, the integrated walkthrough showed that the implemented sensing, local display, physical controls, wireless communication, mobile dashboard, operating-state logic, and emergency software interlock functioned coherently within one combined scenario. The result remains limited to the electronic, firmware, interface, and software-interlock functions implemented in the tested prototype; physical relay or solenoid actuation was not evaluated.

4.5 Questionnaire Survey on the Need for and Acceptance of the Proposed SRSS

4.5.1 Survey Method and Respondent Profile

A supplementary Arabic-language online questionnaire was conducted to examine public experience with respiratory illness, perceived access to respiratory support, acceptance of a proposed smart home respiratory-support concept, preferred system functions, and the patient groups expected to benefit most. The form received 245 submissions. The number of valid answers varied between 230 and 238 because some respondents did not answer every question.

The questionnaire measured public perceptions and expected acceptance of the proposed concept. It was not a controlled hands-on usability evaluation of the developed prototype. Therefore, its results should be interpreted as supplementary evidence of perceived need and interest, rather than as evidence of technical performance, clinical effectiveness, or safety.

Age information was provided by 237 respondents. Most respondents were between 20 and 35 years old (73.0%). Respondents older than 50 years represented 10.5%, those younger than 20 represented 8.9%, and the 36–50 age group represented 7.6%. This age concentration should be considered when interpreting the survey findings.

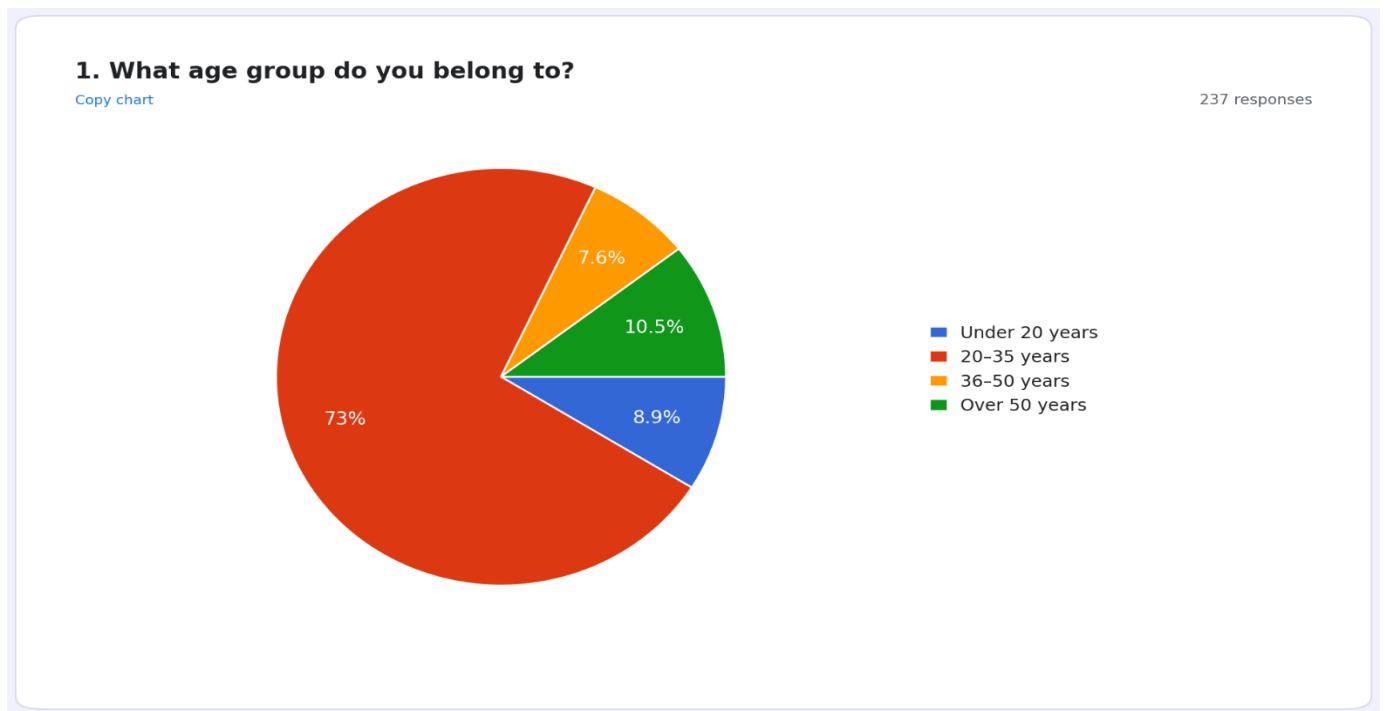


Figure 4.13: Age Distribution of the Questionnaire Respondents

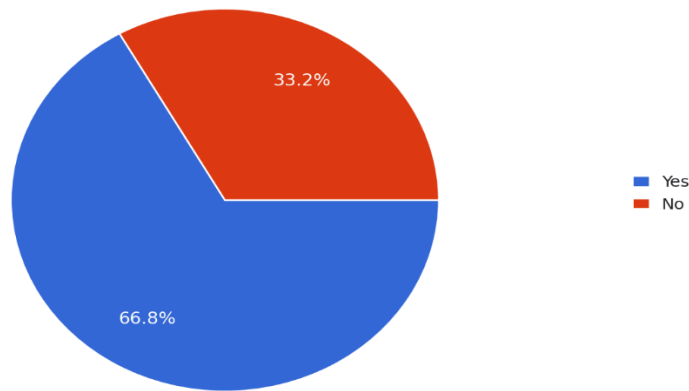
4.5.2 Previous Respiratory Experience

Among 238 respondents, 66.8% reported that they or a family member had previously been infected with COVID-19. In a related question answered by 233 respondents, 63.5% reported that the affected person experienced reduced oxygen saturation or breathing difficulty during the infection. These results show that a large proportion of the surveyed group had direct or family-level experience with respiratory symptoms.

2. Have you or any member of your family ever been infected with Coronavirus (COVID-19)?

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238 responses



3. During the infection, did the patient experience low oxygen saturation or difficulty breathing?

[Copy chart](#)

233 responses

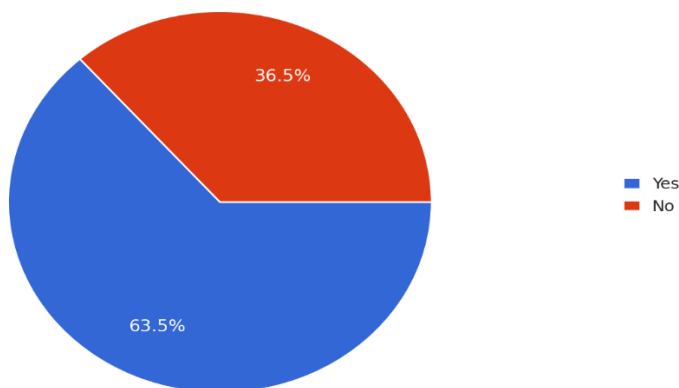


Figure 4.14: Reported COVID-19 Experience and Respiratory Difficulty During Infection

4.5.3 Access to Respiratory Support and Likelihood of Use

Access to a hospital bed or respiratory-support device during the pandemic was evaluated by 237 respondents. Only 7.6% reported that support had been easily available. A further 32.5% reported availability with difficulty or delay, while 22.8% described access as very difficult. The remaining 37.1% stated that they did not need respiratory support.

Acceptance of the proposed home system was also positive. Of the 233 respondents who answered the question, 77.7% stated that they would be likely to use a smart home respiratory-support device if it were recommended by a physician, while 22.3% answered no.

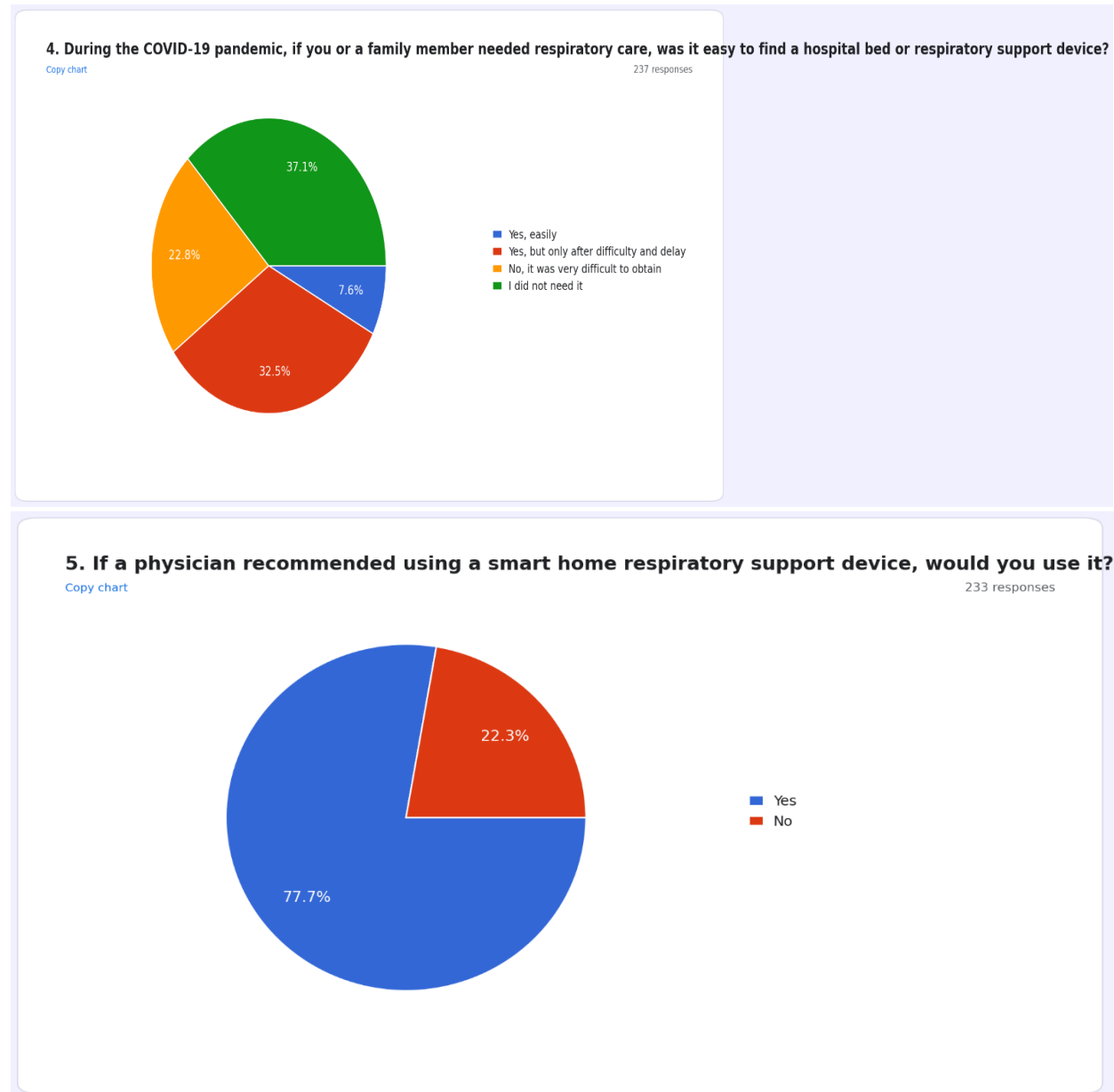


Figure 4.15: Access to Respiratory Support During the Pandemic and Likelihood of Using the Proposed System

4.5.4 Preferred System Functions

Respondents were allowed to select more than one preferred function. Continuous SpO₂ monitoring was the most frequently selected feature, chosen by 191 of 237 respondents (80.6%). Immediate warnings in the event of danger were selected by 180 respondents (75.9%), while heart-rate monitoring was selected by 159 respondents (67.1%). Sending

readings directly to the physician was selected by 143 respondents (60.3%), mobile-application control by 140 respondents (59.1%), and airflow monitoring by 127 respondents (53.6%).

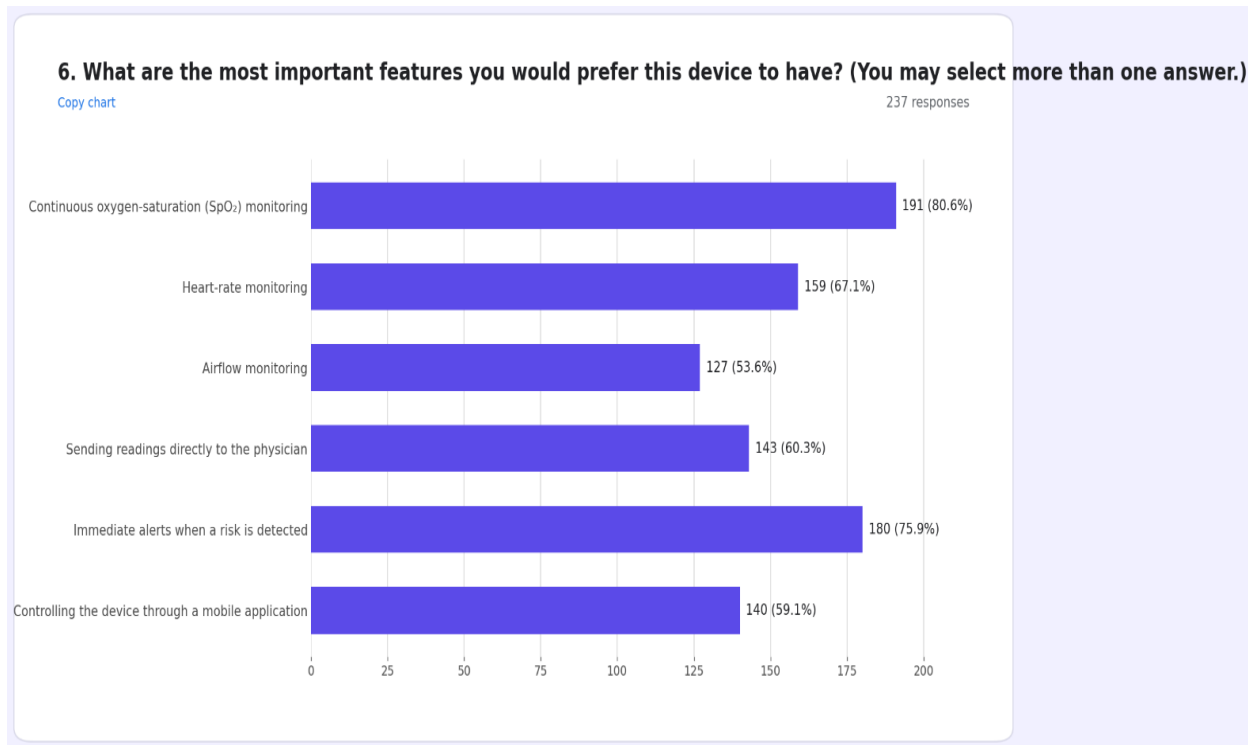


Figure 4.16: Preferred Functions of the Proposed SRSS

4.5.5 Patient Groups Expected to Benefit

The perceived-beneficiary question also allowed multiple selections. Older adults with reduced lung function were selected most frequently, by 177 of 235 respondents (75.3%). Patients with chronic obstructive pulmonary disease were selected by 159 respondents (67.7%), followed by patients with COVID-19 or another chronic lung disease (65.5%), asthma patients (59.1%), and patients requiring temporary respiratory support following surgery (52.3%). Other groups were selected by 15.7%.

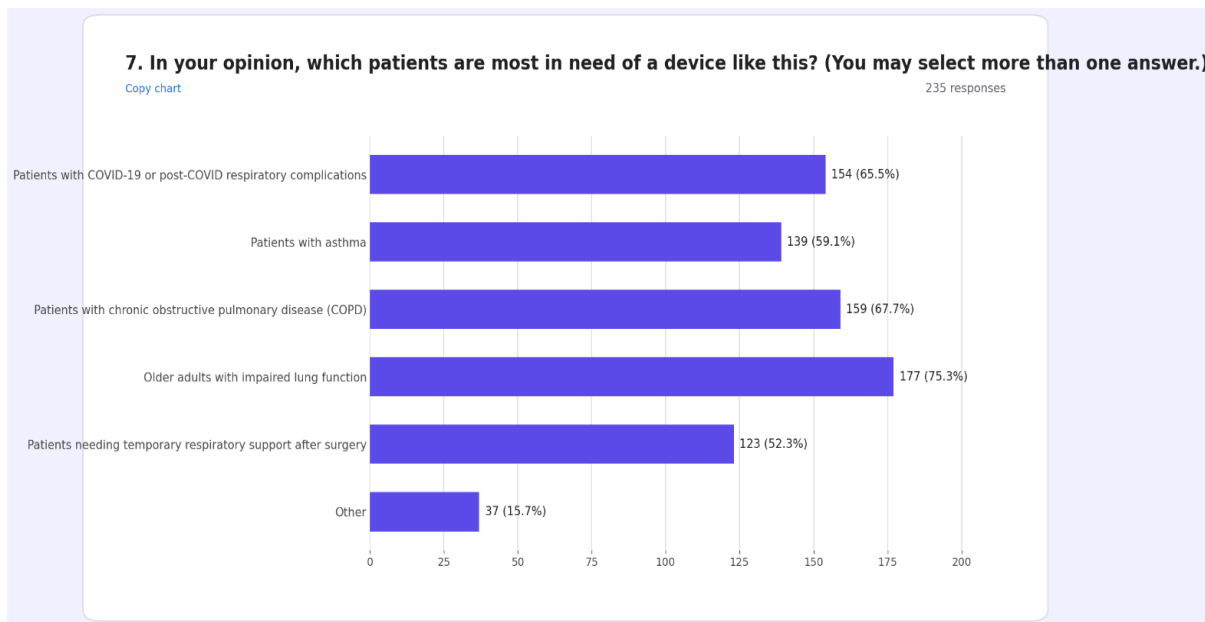


Figure 4.17: Patient Groups Perceived by Respondents to Benefit Most from the Proposed System

4.5.6 Perceived Value of Remote Physician Monitoring

Remote physician monitoring was viewed positively by most respondents. Among 230 valid responses, 50.0% believed that remote follow-up through a mobile application would greatly increase patient safety, and 42.6% believed that it would increase safety to some extent. Only 7.4% did not expect remote monitoring to improve safety. Thus, 92.6% expressed a positive view of the potential safety benefit of remote follow-up.

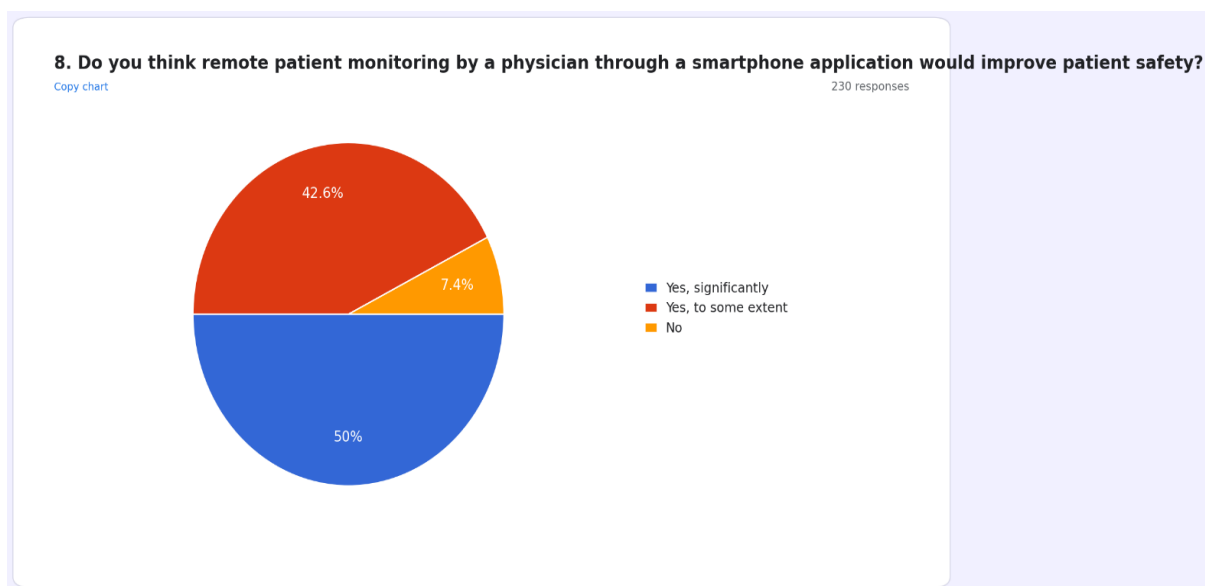


Figure 4.18: Perceived Effect of Remote Physician Monitoring on Patient Safety

4.5.7 Agreement with the Expected Benefits

The agreement-scale question evaluated three expected benefits of the proposed concept: improving home patient monitoring, reducing the need for hospital admission in stable cases, and enabling physicians to monitor patients remotely and intervene earlier when deterioration occurs. As shown in Figure 36, agreement was the dominant response for all three statements, while neutral and disagreement responses were substantially lower.

These responses describe the expectations of the surveyed public and should not be presented as demonstrated clinical outcomes. The current prototype evaluation did not test whether the device reduces hospitalization or improves patient outcomes.

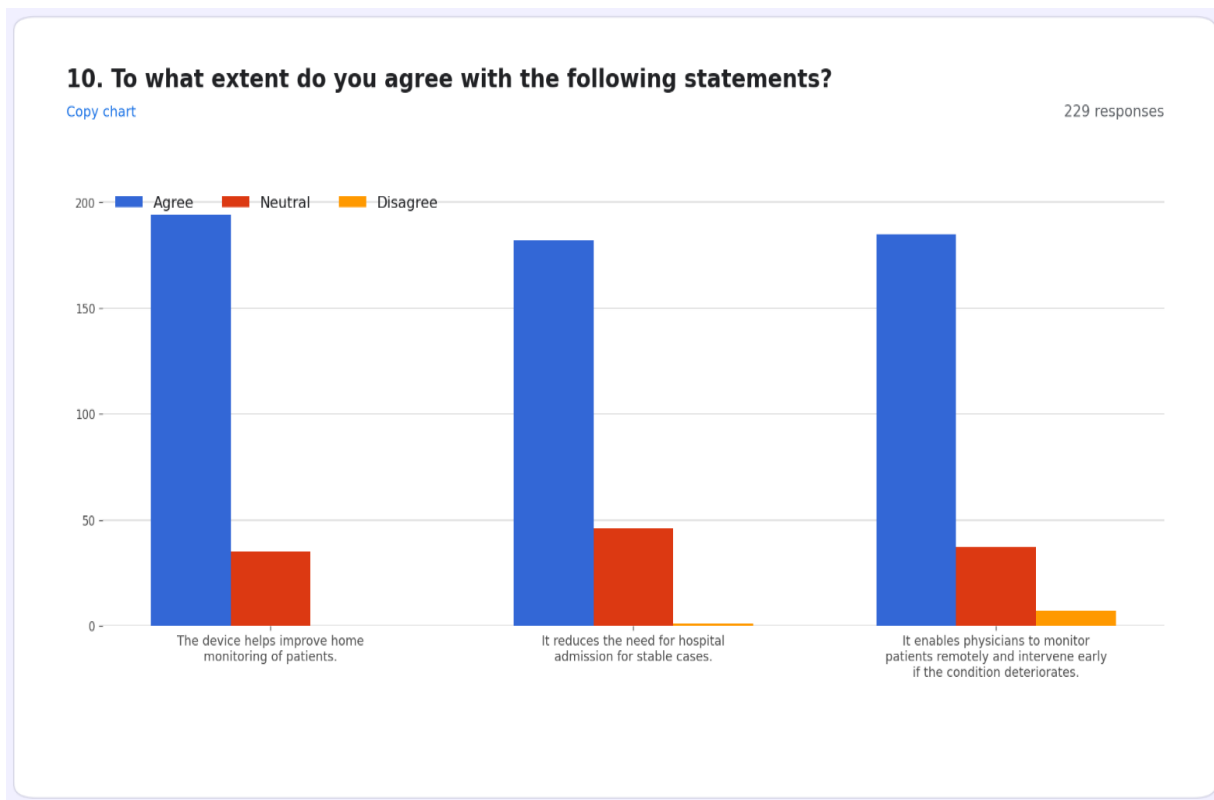


Figure 4.19: Respondents' Agreement with the Expected Benefits of the Proposed SRSS

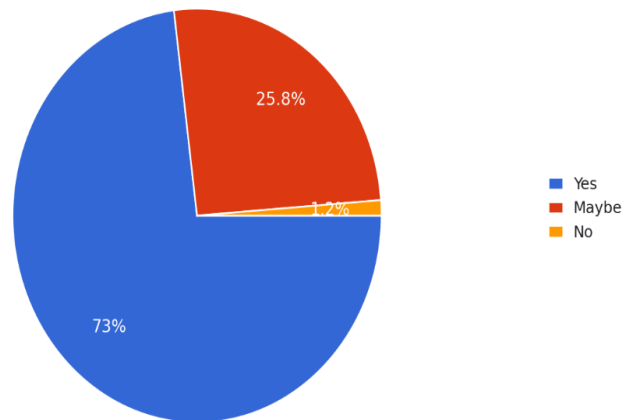
4.5.8 Purchase Intention and Future Interest

Purchase intention was strongly positive. Of 233 respondents, 73.0% stated that they would consider purchasing the device for a family member if a physician recommended it, 25.8% answered maybe, and approximately 1.2% answered no. In addition, 87.7% of 236 respondents expressed interest in learning more about the device after its future introduction, compared with 12.3% who were not interested.

9. If a physician recommended this device for you or a family member, would you consider purchasing it?

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233 responses



11. Would you like to learn more about this device when it becomes available in the future?

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236 responses

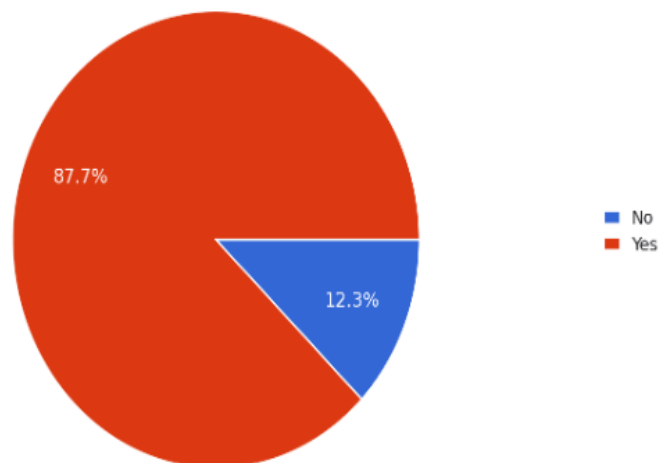


Figure 4.20: Purchase Intention and Interest in Learning More About the Proposed Device

4.5.9 Overall Interpretation and Survey Limitations

Taken together, the questionnaire results indicate substantial public interest in a smart home respiratory-support and monitoring concept. The highest-priority functions were continuous SpO₂ monitoring, immediate risk alerts, heart-rate monitoring, and communication of readings to a physician. These preferences support the monitoring, alarm, wireless-display, and remote follow-up objectives of the SRSS project.

The results also provide a practical basis for future development priorities. Oxygen-saturation monitoring, reliable emergency warnings, secure transmission of readings, and a simple mobile interface should receive particular attention. The responses regarding beneficiary groups suggest that older adults and patients with chronic respiratory conditions are the groups most strongly associated with the proposed system concept.

Several limitations must be considered. The questionnaire used a convenience sample and was dominated by respondents aged 20–35 years. The number of valid answers differed among questions, the responses were self-reported, and no clinical status was independently verified. Most importantly, the respondents evaluated the proposed concept rather than completing a controlled hands-on test of the prototype. The survey therefore supports perceived need and acceptance only; it does not establish usability, measurement accuracy, patient safety, clinical effectiveness, or readiness for medical use.

Table 4.16 summarizes the principal questionnaire findings.

Table 4.16: Summary of the Questionnaire Results

Survey Item	Valid Responses	Main Finding
Total submissions	245	Overall number of submitted forms.
Age distribution	237	73.0% were 20–35 years old.
COVID-19 experience	238	66.8% reported personal or family infection.
Breathing difficulty / low SpO₂	233	63.5% reported respiratory difficulty during infection.
Easy access to respiratory support	237	Only 7.6% reported easy access.
Likely to use if recommended	233	77.7% answered yes.

Remote monitoring and safety	230	92.6% expected a large or moderate safety benefit.
Purchase intention	233	73.0% yes and 25.8% maybe.
Interest in learning more	236	87.7% answered yes.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This project presented the design, implementation, and bench-level evaluation of a Smart Respiratory Support System (SRSS). The main objective was to develop one embedded prototype capable of monitoring selected physiological and respiratory parameters while providing local display, physical control, audible feedback, and direct wireless access. The implemented system was built around the ESP32 controller and integrated the MAX30105 optical sensor, DS18B20 temperature probe, YF-S401 airflow sensor, two I²C LCD modules, four operating-mode buttons, a STOP button, a piezo buzzer, and a locally hosted mobile web interface.

The completed prototype demonstrated the functional integration of its main electronic and software subsystems. Heart rate, oxygen saturation, airflow, and temperature values were acquired and presented on the two LCD modules, while the same information was made available through the mobile interface. ESP32 created a local Wi-Fi access point, allowing a smartphone to connect directly to the system without an external router, cloud service, or internet connection. QR-code onboarding also simplified access to the Wi-Fi network and the local dashboard.

Five synchronized four-parameter measurement sets were retained from adult participants at rest. The recorded mean values were 74.2 BPM for heart rate, 97.8% for SpO₂, 10.72 L/min for airflow, and 36.7 °C for temperature. The local LCDs and the mobile interface showed the same recorded values during the reported trials. The narrow variation observed in SpO₂ and temperature supported stable presentation under the selected test conditions, while the differences in heart rate and airflow reflected the variation between the recorded participant sessions. These results demonstrate functional sensor acquisition and data presentation, but they do not establish calibrated accuracy because the readings were not compared with certified reference instruments.

The local controls and wireless commands were handled through the same internal operating state logic. Monitoring, Nebulizer, Ventilation, and Controlled Ventilation states could be selected from the physical buttons and through HTTP-based commands. The selected state was then reflected on the LCDs and the mobile interface, and the buzzer provided audible confirmation. The integrated walkthrough showed that the sensing, display, control, communication, and feedback functions could operate together in one continuous sequence.

A further project outcome was the successful fabrication and assembly of a dedicated printed circuit board for the control circuitry. The board combined the ESP32 mounting headers, relay section, terminal connectors, and supporting components on one platform. This reduced loose wiring, improved the internal arrangement of the electronics, and provided a more practical basis for mounting the control hardware inside the prototype enclosure.

The STOP function operated as an emergency software interlock. When STOP was activated, the system entered the Stopped state, the buzzer alarm was enabled, and warning messages appeared on the local displays and the mobile interface. Remote commands could not clear the stop

condition, and normal operation resumed only after a physical acknowledgement using the Monitoring button. This behavior confirmed the implemented state-locking logic. However, the relay and solenoid were not controlled by the tested firmware, so the function did not physically interrupt the airflow path.

Overall, the project achieved its main goal of producing a working and integrated SRSS prototype for preliminary bench demonstration. The strongest result was the coordinated operation of the sensing, display, local control, Wi-Fi, mobile-interface, PCB, and software interlock functions within one system. At the same time, the evaluation was limited to one prototype, five retained participant measurement sets, and normal indoor conditions. Sensor calibration, repeated-measurement analysis, exact response-time and power characterization, automatic actuator control, physical airflow shutdown, long-duration reliability, and clinical validation remain outside the completed scope.

The developed SRSS should therefore be considered a functional engineering prototype rather than a finished medical device. It provides a clear foundation for further technical development and demonstrates how the proposed monitoring and control functions can be combined in a compact embedded system.

5.2 Future Work

Future development should focus on improving measurement confidence, completing the electromechanical control path, strengthening safety, and evaluating the system under a wider range of operating conditions. The following activities represent the main recommended directions for the next version of the prototype.

- 1. Sensor calibration and accuracy evaluation:** Each sensing channel should be compared with an appropriate reference instrument. A commercial pulse oximeter, a calibrated thermometer, and a calibrated flow meter can be used to calculate measurement error, bias, and repeatability. The resulting data should then be used to refine the firmware calculations and calibration factors.
- 2. Expanded and repeated testing:** The number of participants and the number of measurements per participant should be increased. Repeated readings should be recorded under stable conditions, and the sensors should also be tested under different placement, motion, ambient-light, temperature, and airflow conditions. This would provide a more reliable description of stability and repeatability.
- 3. Relay and solenoid integration:** The relay and solenoid should be connected to the firmware state machine so that each operating mode produces a defined physical response. The actuator output should be tested separately before being included in a complete system trial. Valve response time, relay rating, electrical isolation, and failure behavior should also be verified.

4. Hardware emergency shutdown: The current STOP function should be extended from a software interlock to a fail-safe hardware shutdown. Activating STOP should place the airflow-control hardware in a safe condition even if the ESP32 or the communication link fails. The design should also detect relay, valve, sensor, and power faults and provide clear audible and visual warnings.

5. Closed-loop airflow and pressure control: A future version should include a suitable pressure sensor and a controllable airflow source. Feedback control could then be used to maintain the required flow or pressure within defined limits. This stage should first be tested on a controlled bench setup or lung simulator before any medically supervised evaluation is considered.

6. Data logging and mobile-interface development: The mobile interface can be improved by adding time-stamped data storage, trend graphs, alarm history, connection status, and clear sensor-quality indicators. Separate monitoring and control permissions should be considered, and confirmation should be required before safety-related commands are executed.

7. Power, PCB, and enclosure improvement: Operating current, peak current, electrical power, and battery runtime should be measured using suitable instruments. A rechargeable backup supply and low-battery warning can then be added if portable operation is required. The PCB should also be revised to include protection, fusing, labeled connectors, test points, and improved isolation, while the enclosure should provide safer cable routing and easier maintenance access.

8. Wireless security and reliability testing: The fixed Wi-Fi credentials and unencrypted local HTTP communication should be replaced by a more secure access method. Future tests should measure connection range, reconnection behavior, command latency, wall penetration, and the effect of multiple connected devices. The system should also recover safely after temporary communication loss.

9. Long-duration and safety validation: The prototype should be operated continuously for extended periods to identify thermal, memory, sensor, communication, and power-related problems. Formal risk analysis, alarm verification, electrical-safety checks, and usability testing would be required before any patient-related work. Clinical evaluation should only be considered after technical safety has been demonstrated and the necessary medical, ethical, and regulatory approvals have been obtained.

The immediate priorities are calibrated sensor comparison, repeated measurement testing, and safe firmware control of the relay and solenoid. Completing these tasks would address the most important limitations identified in Chapter 4 and move the SRSS from a functional demonstration toward a more reliable and safety-oriented engineering system.